

Financial cycles: Characterisation and real-time measurement[☆]Yves S. Schüler^{a,*}, Paul P. Hiebert^b, Tuomas A. Peltonen^c^a Deutsche Bundesbank, Germany^b European Central Bank, Germany^c European Systemic Risk Board, Germany

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ABSTRACT

We demonstrate that financial cycles (identified as the common fluctuation in credit and asset prices, proxying balance-sheet leverage) differ across G-7 countries in terms of duration. This contradicts a similar-duration assumption inherent in the Basel III credit-to-GDP gap guiding countercyclical capital buffers. Against this backdrop, we propose an empirical methodology for constructing country-specific financial cycles that relaxes the similar-duration assumption and is based on the common fluctuation of several variables. Using credit and asset prices as inputs to our methodology, we show that constructed financial cycles significantly outperform the Basel III credit-to-GDP gap in predicting financial crises.

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1. Introduction

Accurate measurement of financial cycles is critical to underpin countercyclical macroprudential policy mandates around the world, not least to inform the use of the countercyclical capital buffer (CCyB) legislated in 73 countries.¹ Benchmark

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¹ For a detailed overview of macroprudential policy tools and the role of financial cycles, see [Constâncio et al. \(2019\)](#), [Alam et al. \(2019\)](#), [Claessens \(2015\)](#).

methods applied to measure financial cycles employ the Hodrick and Prescott (1981, 1997) filter (HP filter) as prescribed in Basel III legislation (see Basel Committee on Banking Supervision, 2010), or a related specification of the band pass filter.² However, these approaches suffer from two shortcomings. First, they assume a similar duration of financial cycles across countries and time. This may give rise to spurious cycles in widely employed proxies of the financial cycle, such as the Basel III credit-to-GDP gap (see Harvey and Jaeger, 1993; Cogley and Nason, 1995; A'Hearn and Woitek, 2001; Schüler, 2018).³ Spurious cycles may neither cater for country specificities of financial crises, nor an adequate setting of the CCyB. Second, the statistical filtering methods embed endpoint biases, thereby limiting real-time policy use (see Edge and Meisenzahl, 2011). Unreliable real-time properties of filters can lead to large ex post revisions of financial cycle measures and, thus, add uncertainty to the real-time calibration of macroprudential policies.

This study addresses both shortcomings. Based on country-specific financial cycle characteristics (that challenge the similar-duration assumption), we propose an empirical methodology to construct composite financial cycle indicators exploiting the common fluctuation of several indicators. Our methodology (i) does not impose similar durations across countries or across time and (ii) is suited for real-time applications. Using credit and asset prices as inputs to our approach, we show that constructed financial cycles significantly outperform the Basel III credit-to-GDP gap in predicting systemic banking crises.

We adopt the concept of leverage cycles of Geanakoplos (2010) for identifying financial cycles. Specifically, we identify financial cycles via a so-called “balance sheet” channel. That is, one implication of credit market frictions is that the state of balance sheets is an important determinant of agents' ability to borrow and lend (e.g. Bernanke and Gertler, 1999). Changes in asset prices alter an agent's net worth, which influences the scale of borrowing and lending. Based on this idea, we measure financial cycles as the common movement of credit and asset prices (consistent with Claessens et al., 2012; Jordà et al., 2015b). Common expansions and contractions of credit and asset prices proxy the strength of balance sheets, with implications for excess aggregate credit growth. In our application, we consider the co-movement of total credit to the private non-financial sector with asset price aggregates across all main asset markets – that is, house prices, equity prices, and corporate bond prices for G-7 countries from 1970Q1 until 2018Q4. To benchmark financial cycle characteristics, we construct analogous composite business cycles, capturing the co-movement of GDP, consumption, investment, and hours worked.

Our proposed methodology isolates country-specific financial cycle frequencies through *power cohesion*. Power cohesion is suited for identifying cycles common to a set of variables that strongly contribute to their overall co-variance. More formally, we define power cohesion as the linear combination of normalised absolute cross-spectra for all pairwise combinations of variables in the set considered. Analysing a narrow financial cycle (credit and house prices) and a broad financial cycle (narrow cycle plus equity and bond prices), we find that financial cycle durations vary greatly across countries, with the average being 15.6 years and 14.2 years respectively. At one extreme, Germany's financial cycle is the shortest of the G-7 countries, closely corresponding with a business cycle lasting around 6.6 years. At the other extreme, Japan's financial cycle is long and closely corresponds to a protracted business cycle (lasting over 20 years) that relates to the “lost decades”. Overall, we observe that financial cycles are longer and have a larger amplitude than business cycles, with business cycles lasting only around 9 years on average (or 6.6 years excluding Japan). Greater differences in the duration of financial cycles relative to business cycles is found for countries with higher home ownership rates, such as Italy, the UK, or the US.

Building on power cohesion, we construct composite financial cycles by combining standardised quarterly growth rates of credit and asset prices. The quarterly growth rates allow for different cycle durations across countries as well as time. We show that merely linearly combining variables yields clear cyclical movements, as common fluctuations are emphasised and idiosyncratic ones are dampened. These composite financial cycles synchronise strongly across countries for most of the G-7 economies. The financial cycles of Germany and Japan are distinct in this respect – being weakly related among G-7 countries with very distinct movements in credit and house prices (Hume and Sentance, 2009; André, 2010; Knoll et al., 2017). Business cycles, in contrast, relate homogeneously across G-7 countries. We evaluate financial cycles through different predictive signalling exercises for financial crises. Combining information into a composite financial cycle outperforms – especially in real time – the input indicators, as well as the credit-to-GDP gap. The exercise suggests that a broad set of asset prices – possibly due to their forward-looking nature – may be crucial for the real-time prediction of financial crises.

Our findings support studies, which have stressed a higher length, amplitude, and degree of symmetry in financial cycles compared with business cycle counterparts. Claessens et al. (2012) find that credit, house price, and equity price cycles are longer and more volatile than business cycles identified through GDP. Aikman et al. (2015) observe an important medium-term dimension in credit cycles, distinct from business cycles. Drehmann et al. (2012) and Borio (2014) argue, as well, that medium-term variance is especially important for credit cycles as well as house price cycles, but not for equity prices. Furthermore, the authors note that the length of financial cycles increased during the Great Moderation. Hiebert et al. (2018) provide evidence that financial cycles are far more symmetric than business cycles. Other studies reinforce this evidence, applying a range of methodologies such as indirect spectral analysis (Strohsal et al., 2015a,b), multivariate unobserved

² To date, a series of studies has applied the Basel III specification of the HP filter (or a related specification of the band pass filter) for constructing alternative proxies of financial cycles based on credit, asset prices, or combinations thereof. See Drehmann et al. (2012), Borio (2014), Stremmel (2015), Anundson et al. (2016), Bauer and Granziere (2017), Samarina et al. (2017), Meller and Metiu (2017), Detken et al. (2018), Jordà et al. (forthcoming).

³ Schüler (2018) observes that the filtering methodology – the Hodrick and Prescott (1981, 1997) filter with a smoothing parameter of 400,000 (but also related band pass filters) – means that the credit-to-GDP gaps of G-7 countries are biased towards a 30-year cycle duration. This also holds true for similarly constructed gaps derived from credit, house prices, equity prices, bond prices, or GDP.

component models (Rünstler and Vlekke, 2018; Galati et al., 2016), or wavelet techniques (Verona, 2016; Mandler and Scharnagl, 2018).

The present study also relates to the literature on business cycle measurement and characteristics. Our findings suggest a medium-term component in business cycles, corroborating the findings of earlier work (e.g. A'Hearn and Woitek, 2001; Comin and Gertler, 2006). For the US, our methodology consistently estimates the business cycle length as derived by the NBER dating committee, while also indicating important fluctuations above eight years. Specifically, an estimated frequency band for US business cycles (derived from PCoh) covers cycles lasting between 2.6 and 15.4 years. This is in line with recent discussions on revising estimates of potential output after the global financial crisis (Phillips and Jin, 2015; Coibion et al., 2018).

The remainder of the paper is organised as follows. Section 2 discusses the indicators that feed into our financial cycle measurement. Section 3 introduces our power cohesion methodology to extract common cyclical movements of a set of cycle indicators, and presents the characteristics of financial and business cycles across G-7 countries. Section 4 suggests an approach to extract common movements of indicators in the time domain, which are then used to shed light on financial cycle synchronisation versus business cycle synchronisation and the prediction of past systemic banking crises and their preceding vulnerability periods. Section 5 concludes.

2. Financial cycle indicators, business cycle indicators, and empirical issues

Financial cycle indicators.

Much the same way as business cycles relate to recessions, financial cycle indicators should relate to financial recessions, i.e. recessions that occur in tandem with banking crises.⁴ In contrast to standard recessions, financial recessions occur less frequently, are deeper, and last longer. Further, financial recessions follow credit booms (see Jordà et al., 2013; Boissay et al., 2016).

In this light, credit is a *necessary* indicator for financial cycles. Boissay et al. (2016) outline several empirical facts about credit, showing that it follows distinct movements such as higher variance and persistence around financial recessions than other types of recessions. Schularick and Taylor (2012) provide evidence that lagged credit growth predicts financial crises well, underscoring how credit is essential to the build-up of imbalances presaging financial recessions and their subsequent unwinding. Several studies argue that credit can be a source of instability, and not just an amplifier of shocks as described in the financial accelerator approach (see Bernanke, 1983; Gertler, 1988; Kiyotaki and Moore, 1997; Bernanke et al., 1999). In his seminal work, Minsky (1977) argues that there may be endogenous financial instability through credit expansions, which push investor sentiment into interchangeably overoptimism or fear.⁵

While credit may be a necessary element of financial cycles, it does not seem *sufficient* to capture financial cycles. Mendoza and Terrones (2008), Gorton and Ordoñez (2016) note that not all credit booms lead to a financial crisis. In this regard, asset prices complement credit and complete financial cycles via a so-called “balance sheet” channel. One implication of credit market frictions is that the state of balance sheets is an important determinant of agents' ability to borrow and lend (e.g. Bernanke and Gertler, 1999). Changes in asset prices alter an agent's net worth, which influences the scale of borrowing and lending. The importance of this interaction has been the centre of studies on leverage cycles as by, e.g. Geanakoplos and Fostel (2008), Geanakoplos (2010), as well as work by Gromb and Vayanos (2002), Brunnermeier and Pedersen (2009), and Adrian and Shin (2010).

We capture the balance sheet channel by combining credit in a financial cycle with house, equity, and bond prices, i.e. the three largest asset classes that, moreover, are available for a long time span required to characterise medium-term aspects of financial cycles. The importance of *real estate* is noted by the likes of Iacoviello (2005), who introduces collateral constraints governed by real estate values into a monetary business cycle model, as empirics indicate that a large part of borrowing is secured by real estate (see also Campell and Hercowitz, 2009; Hartmann, 2014; Justiano et al., 2015). Jordà et al. (2015a, 2016) note that risks to financial stability are strongly linked to real estate lending developments. A role for equity prices is noted by Claessens et al. (2012), Claessens et al. (2011), who provide empirical evidence that recessions marked by *equity* together with *house price* busts tend to be deeper and longer. Additionally, Jordà et al. (2015b) report that most build-ups of systemic risk involved both equity and house prices post-World War II. Corporate *bond prices* have been held to be an important indicator of the soundness of the financial system, e.g. during the global financial crisis (Gilchrist and Zakrajšek, 2012). Gilchrist et al. (2009) find that an unexpected increase in bond spreads, i.e. corporate yield minus risk free yield, cause large and persistent contractions in economic activity. Further, bond market information is an integral component of financial stress indices. Shocks to such indices have been shown to be highly detrimental to the economic outlook (e.g. Hubrich and Tetlow, 2015; Fink and Schüler, 2015).⁶

⁴ Financial recessions are defined in this way by Jordà et al. (2013).

⁵ Bhattacharya et al. (2015) model Minsky's hypothesis formally with the possibility of endogenous default, with support for cycles, in which agents update their expectations during good times and increase their leverage accordingly.

⁶ Given discussions on the validity of asset prices prior to the global financial crisis, it is important to stress that our approach considers the joint movement of credit with a complete portfolio of asset prices. First, the combination of variables dampens idiosyncratic movements, such as erratic asset price valuations. Second, we are precisely interested in emphasising asset price movements that go in tandem with credit developments (in the spirit of Geanakoplos, 2010) and that possibly do not reflect fundamental values.

Building on this rich literature, we opt for a *narrow* and *broad* measure of a financial cycle. The *narrow* measure, incorporating credit and house prices, is motivated by the close relationship between credit and real estate as argued by [Iacoviello \(2005\)](#), [Jordà et al. \(2015a, 2016\)](#), or their statistical properties (such as strong contemporaneous co-movement) as discussed at length in [Appendix A.2](#). The *broad* measure, also including financial asset markets through equity and bond prices, is motivated by evidence that a larger set of asset prices might be vital to identify emerging risks, including on a global scale (see [Rey, 2015](#); [Miranda-Agrippino and Rey, 2015](#); [Breitung and Eickmeier, 2014](#)).

Business cycle indicators.

Besides indicators relevant for financial cycle measurement, we use a set of indicators to measure business cycles. Analysing these indicators with the same methods allows us to contrast financial cycle facts with business cycle facts.

Business cycle measurement has a long history, with early attempts to establish a set of business cycle regularities, for instance, by [Burns and Mitchell \(1946\)](#). One of these business cycle regularities is that output, consumption, investment, and hours worked display strong positive contemporaneous correlation (see [King and Rebelo, 1999](#)). Real business cycle models actually embed this fact, such as the seminal model presented by [Kydland and Prescott \(1982\)](#). We build on this research programme and measure business cycles through the co-movement of the same four variables – output, consumption, investment, and hours worked.

Empirical issues.

We analyse cycles in growth rates, rather than classical cycles or growth cycles ([Harding and Pagan, 2005](#)), among others, as the spectral approach taken in this paper requires stationary processes. In doing so, we depart from common financial cycle extraction methods, such as those specified for the Basel III credit-to-GDP gap defined as a growth cycle. This is an important departure, because cycles in growth rates are not subject to the problem of spurious cycles or end-point biases (see [Schüler, 2018](#); [Edge and Meisenzahl, 2011](#)). That is, cycles in growth rates allow for a changing duration of cyclical fluctuations. Methods designed to extract growth cycles from typical economic time series – e.g. HP or band pass filters – induce a certain cycle duration, depending on the filter's specification (such as the choice of the HP smoothing parameter).⁷ [Schüler \(2018\)](#) shows that the Basel III specification of the HP filter induces, by using a smoothing parameter of 400,000, longer duration cycles that last up to 30 years.

To preserve cyclical turning points, a quarter-on-quarter differencing is implemented. This is in keeping with the well-established NBER methodology (and CEPR in Europe) for identifying peaks and troughs of the business cycle on the basis of quarterly changes in GDP. Furthermore, similar to [Comin and Gertler \(2006\)](#) or [Stock and Watson \(2005\)](#), we purge longer-term cycles from our indicators. Longer-term cycles are not subject to our analysis as they would bias the correlations between indicators that enter our estimated spectral densities, and are irrelevant for policy making. Specifically, we remove cycles that are longer than 50 years from the growth rates of indicators using the [Christiano and Fitzgerald \(2003\)](#) band pass filter.⁸ Only within our signalling exercise (Section 4.4), for which we construct real-time indices, do we use the unfiltered growth rates to avoid end-point biases.

In sum, total credit (Δcr), house prices (Δp_h), equity prices (Δp_e), output (Δq), consumption (Δco), investment (Δi), and hours worked (Δh), spanning from 1970Q1 until 2018Q4, are measured in quarterly filtered growth rates excluding cycles of duration longer than 50 years. Unfiltered growth rates are used in the signalling exercise. Corporate bond yields are transformed to reflect filtered growth in bond prices, to be in line with the interpretation of house and equity prices, by exploiting the relationship: $p_{b,t} = 1 / (1 + y_{b,t})$, where $p_{b,t}$ denotes the price and $y_{b,t}$ the current yield of the respective bond at time t . As we analyse bond indices – given that we require a long history – we do not have all the information available for this transformation. Specifically, we assume that all yields have been transformed to zero coupons. Further, given data limitations for the set of countries considered, we resort to a simplifying assumption that indices are derived from a portfolio of bonds with constant average yield to maturity. Finally, all variables are in real terms, deflated by a corresponding consumer price index. The exact details on the variables are provided in [Appendix A.1](#).

3. Financial cycle frequencies

This section investigates whether financial cycle indicators co-move at different cycle durations across countries or when compared to the co-movement of business cycle indicators. We first present a methodology for this purpose and then report the results.

3.1. Methodology: Power cohesion – discovering important cycle frequencies across a set of indicators

For identifying financial cycle frequencies common to a set of variables, we propose a novel multivariate spectral measure that we call power cohesion (PCoh). PCoh allows for an intuitive description and analysis of the cyclical structure of fluctuations that is common to a set of indicators. The measure quantifies the co-movement (similar to correlation) of possibly

⁷ Typical economic time series can be argued to be best approximated by a random walk; see [Hamilton \(in press\)](#).

⁸ Here, the procedure is not subject to the problem of spurious cycles because we apply the filter to growth rates. Spurious cycles emerge when applying the filter to the level of a difference stationary time series, i.e. when extracting growth cycles. Furthermore, note that this procedure is similar to removing longer-term cycles from growth rates via a local mean model as done by [Stock and Watson \(2005\)](#). We prefer the use of a band pass filter over a local mean model, as it allows us to exactly articulate the frequencies that we do not want to consider. A comparison to this approach is available from the authors upon request.

many indicators. It allows us to describe at which cycle frequency (e.g. eight years or so) there is co-movement between indicators that, moreover, explains large parts of the indicators' common variance.

PCoh relates, but is not equal, to well-known quantities that have been long used in business cycle analysis, such as squared coherency (for an overview, see A'Hearn and Woitek, 2001). Furthermore, PCoh relates to Croux et al. (2001), who analyse cyclical similarities across a set of indicators in the frequency domain and name it cohesion. To put our measure into the overall context, we compare PCoh to cohesion based on squared coherency in Appendix A.3 (see Proposition 3). In brief, cohesion based on squared coherency indicates similar cycles irrespective of their importance for overall co-variance. In contrast, PCoh gives greater importance to similar cycles that explain a larger share of co-variance.

More formally, we define PCoh as the linear combination of normalised absolute cross-spectra for all pairwise combinations of variables in the set considered. Let $M \geq 2$ reflect the number of variables, $t = 1, \dots, T$ the time dimension, $X = (X'_1, \dots, X'_T)'$ a $T \times M$ matrix containing stationary stochastic processes with well-defined normalised cross-spectral densities $f_{x_i x_j}(\omega)$, where $X_t = (x_{1,t}, \dots, x_{M,t})$ ($1 \times M$), $1 \leq i \leq M$, $1 \leq j \leq M$, and $\omega \in [-\pi, \pi]$ denoting the respective cycle frequency. PCoh can then be formulated as

$$\text{PCoh}_X(\omega) = \frac{1}{(M-1)M} \sum_{i \neq j} |f_{x_i x_j}(\omega)|, \quad (1)$$

with

$$f_{x_i x_j}(\omega) = \frac{s_{x_i x_j}(\omega)}{\sigma_{x_i} \sigma_{x_j}} = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \frac{\text{Cov}[x_{i,t}, x_{j,t+k}]}{\sigma_{x_i} \sigma_{x_j}} e^{-ik\omega}, \quad (2)$$

where σ_{x_i} and σ_{x_j} are the standard deviations and $s_{x_i x_j}(\omega)$ is the cross-spectrum of $x_{i,t}$ and $x_{j,t}$. $\text{Cov}[x_{i,t}, x_{j,t+k}]$ reflects the cross-covariance between $x_{i,t}$ and $x_{j,t+k}$ and i is the imaginary unit.

Below, in Proposition 1–3, we summarise the characteristics of PCoh. For the relevant proofs, please see Appendix A.3.

Proposition 1. *Considering the absolute values of the cross-spectra implies that PCoh discards phase shifts between variables.*

Proposition 1 means that PCoh neglects lead-lag relations between indicators, i.e. detects common fluctuations, even if a cycle of similar duration is shifted in phase. A formal analysis suggests that, especially for the broad set of financial cycle indicators, there might exist phase shifts between variables; not so for the narrow set, in which case this assumption could be relaxed (see Appendix A.2).

Proposition 2. *The normalisation of absolute cross-spectra by the respective standard deviations (see Eq. (2)) implies that the support of PCoh is normalised. It ranges, when integrated from $-\pi$ to π , from 0 to 1.*

Proposition 3. *PCoh indicates the contribution of different cycle durations to the overall co-variance between indicators. Thus, it differs from the more common measure based on squared coherency that indicates similarities of cycles irrespective of their importance, i.e. irrespective of their contribution to overall co-variance.*

Since we estimate the cross-spectral densities using standard methods, we refer the reader to Appendix A.4 for details on our approach.

3.2. Cross-spectral densities: The building block of power cohesion

Cross-spectral densities, i.e. $s_{x_i x_j}(\omega)$, are one of the building blocks of PCoh. In this section, we consider these in more detail in order to analyse which indicator pairs share common cycles at which duration and, furthermore, to study the amplitude of common fluctuations between indicators. PCoh does not allow for such a comparison due to the normalisation of cross-spectral densities.

In Fig. 1, we show the absolute cross-spectral densities of the US financial and business cycle indicators, measuring covariances in and out of phase (see Appendix A.3 for more details). The other country cases are depicted in Appendix A.5. On the x-axis the graphs indicate the cycle length (from 1.25 years to 50 years). We mark the region of frequencies that are commonly attributed to the business cycle (2–8 years) and the range of frequencies from 8 to 20 years that we find to be important for financial cycles.

Concerning amplitude and in the case of the US, it is evident that financial cycles (Fig. 1a) are more volatile on average relative to business cycles (Fig. 1b). The peaks of the different financial cycle cross-spectral densities are well above the peaks of the cross-spectral densities of business cycle indicators, except for the cases of house and bond prices as well as credit and bond prices, which are, however, still above most of the cross-spectral densities of business cycle indicators. Looking beyond the US at other G-7 countries, we find that Germany's financial cycle has the lowest amplitude – contrasting with strong amplitude in both Italy and the UK.

With respect to duration, US financial cycle indicators share relatively more cyclical variance at medium-term frequencies than business cycle indicators. This can be seen by the relatively elevated values of the cross-spectral densities of financial

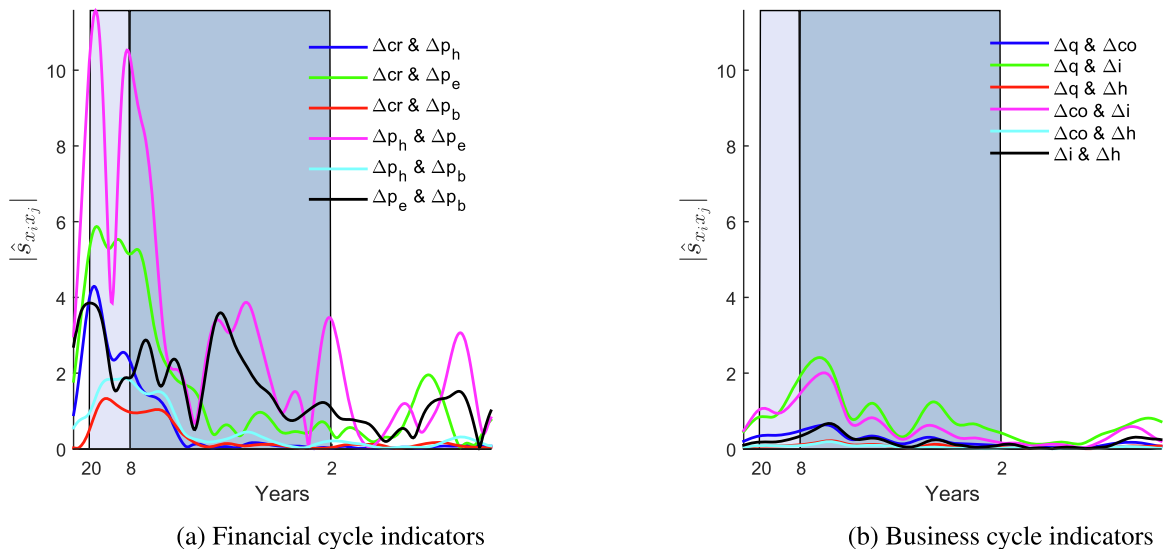


Fig. 1. US: Absolute cross-spectra. *Notes:* This panel shows the absolute cross-spectra of the financial and business cycle indicators. The x-axis measures the frequencies of cycles from 1.25 to 50 years. The blue area depicts business cycle frequencies, i.e. cycles with durations of 2–8 years and the purple area marks frequencies important for financial cycles (8–20 years). Δcr refers to percentage changes in total credit, Δp_h to percentage changes in house prices, Δp_e to percentage changes in equity prices, Δp_b to percent changes in bond prices, Δq to percentage changes in GDP, Δco to percentage changes in consumption, Δi to percentage changes in investment, and Δh to percentage changes in hours worked. For figures to appear in color, please see web version of this article.

cycle indicators within the purple shaded area that compares with relatively lower levels of the cross-spectral densities of business cycle indicators. That is, most important common cycles for business cycle indicators are rather within the short-term region of cycle length (blue shaded area), a feature on the financial side shared only by equity and bond prices.

Common cycles of house and equity prices (Δp_h & Δp_e , pink line) are most pronounced just around 8 and below 20 years. Common cycles of credit and equity prices (Δcr & Δp_e , green line) are most pronounced for all medium-term cycle frequencies, with small spikes at the regions of peaks in house and equity prices, while credit and house prices (Δcr & Δp_h , blue line) share most important frequencies towards 20 years only. Equity and bond prices (Δp_e & Δp_b , black line), similarly to credit and house prices, have their most important cycles around 20 years, while, however, important cyclical variance is also within the short-term region. House and bond prices (Δp_h & Δp_b , turquoise line) as well as credit and bond prices (Δcr & Δp_b , red line) have their most important cycles above 8 years.

For the business cycle indicators the picture is very clear: all pairs of indicators peak around the same frequency region, which is below but close to 8 years. Further, business cycle indicators do not have relatively important cycles located in the medium-term region when compared with financial cycle indicators.

While these results of financial and business cycle indicators seem to hold for most other G-7 countries, there are exceptional cases. Germany and Japan, in particular, differ from other G-7 countries. The common frequency mass for business cycle indicators in Japan appears to reveal historically a much longer duration than for other G-7 countries. On the other hand, German financial cycle frequencies appear to be rather located around business cycle frequencies. Next, we carve out such differences using PCoh.

3.3. Financial cycle frequencies

This section presents PCoh for the narrow (Δcr , Δp_h) and broad (narrow + Δp_e , Δp_b) financial cycle as well as the business cycle. Fig. 2 depicts the case of the US. All other country cases are presented in Appendix A.5. Again, graphs indicate on the x-axis the cycle length from 1.25 to 50 years. The region commonly associated with business cycle frequencies is shaded in blue, while the region in purple represents medium-term frequencies.

To benchmark PCoh as a new approach to measure important cycles pertinent to a set of indicators, we reference the result for US business cycle indicators to recessions as published by the NBER dating committee, which implicitly demarcates business cycle frequencies. To this end, the peak of PCoh for the US business cycle is located at a cycle length of 5.7 years, being identical in length to the reference NBER statistic, also at 5.7 years (average business cycle duration from 1945 to 2009; measuring peak to peak).

For the US, both the broad and narrow measures of financial cycles clearly have most mass located at medium-term frequencies (peak at 15.0 years and 16.9 years, respectively), while that for business cycles is located just below these cycle lengths. While the broad and narrow definitions of financial cycles share similarly important cycle frequencies, the broad

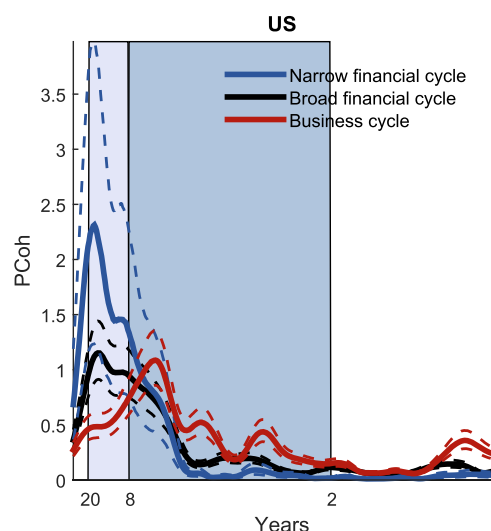


Fig. 2. US: Power cohesion. *Notes:* This graph shows PCoh of the broad (black lines) and narrow financial cycle (blue lines) as well as the business cycle (red lines). Broad refers to the inclusion of all indicators, i.e. credit, house prices, equity prices, and bond prices, whereas narrow is defined by house prices and credit only. The dashed lines indicate the 68% bootstrapped confidence intervals as discussed in [Appendix A.4](#). The x-axis measures the frequencies of cycles from 1.25 to 50 years. The blue area depicts business cycle frequencies, i.e. cycles with durations of 2–8 years, and the purple area (8–20 years) marks frequencies most important for financial cycles. For figure to appear in color, please see web version of this article.

measure envelops relatively more frequency mass amid medium- and short-term frequencies. That said, most cohesion of financial cycle indicators is located in the frequency range within 8 to 20 years. Thus, financial cycle cohesion is above that of business cycles. In contrast, financial cycle indicators do not exhibit strong cohesion for cycles between 2 and 8 years, mostly being below the cohesion of business cycle indicators.

Highest density regions.

Across G-7 countries, PCoh can be conveniently summarised by a highest density chart that depicts 25% of most important frequencies between 1.25 and 50 years (see [Fig. 3](#)). For the financial cycle, we show the narrow (blue) and broad (black) measure, as well as the narrow plus equity prices (turquoise) and narrow plus bond prices (purple) for robustness. Coloured bars indicate at which cycle lengths 25% of most important cohesion between indicators is located, and white dashes indicate the peak of PCoh that is the most important common cycle.

Comparing financial cycle frequencies across countries, we find that for the narrow set of indicators almost all G-7 countries have their most important cycles close to 20 years, albeit with heterogeneity in the exact location of the highest density regions. Here, Germany remains a notable exception, with a highest density region around 8 years cycle duration. Highest density regions for the broad financial cycle are located in the medium-term region as well, being broadly in line with the results for the narrow financial cycle. Interestingly, in the case of Canada, Japan, the UK, and the US there are two separate highest density regions, with the second one around or lower than an 8-year cycle duration. The robustness measures of the financial cycle, i.e. narrow plus one further asset price series, indicate that this result is due to common fluctuations in equity and bond prices. The separate highest density regions disappear with the robustness measures. Only in the case of Japan are the separate highest density regions a characteristic of all financial cycle definitions, possibly relating to the distinct history that includes a substantial cycle build-up followed by the so-called “lost decade” during a lengthy correction phase. This points towards a change in the importance of specific cycle durations across time.

Comparing the financial and business cycle within countries, the following can be noticed: In every country case, a financial cycles’ highest density region covers longer-duration cycles than its business cycle counterpart, the only exceptions being Germany and Japan. Furthermore and across all countries, highest density regions of business cycles include cycles of 8-year duration or shorter. This is not the case for financial cycles.

Why do Germany and Japan reflect exceptions? As indicated, in the case of Japan the long duration of financial and business cycles can be argued to relate to the “lost decades”, referring to the fact that Japan has been mired in stagnation since the 1990s, which may be traced back to the bursting of a bubble in real estate and stock prices in the 1980s (see [Nakaso, 2001](#)). Explanations of the lost decades range from inadequate fiscal policy, the liquidity trap and deleveraging to problems with financial intermediation (see [Hayashi and Prescott, 2002](#); [Hoshi and Kashyap, 2004](#)). The link of the “lost decades” to the financial sector is evident and provides an argument for why financial and business cycle may be of similar duration. With respect to the German results, [Rünstler and Vlekke \(2018\)](#) argue that the duration of financial cycles correlates positively with the private home ownership rate. Indeed, Germany ranks among the countries with the lowest home ownership rate, which, following [Rünstler and Vlekke \(2018\)](#), predicts a short financial-cycle duration. Speaking against such a relation, however, lies our finding of the long-duration financial cycle of Japan, because Japan likewise ranks among the countries of

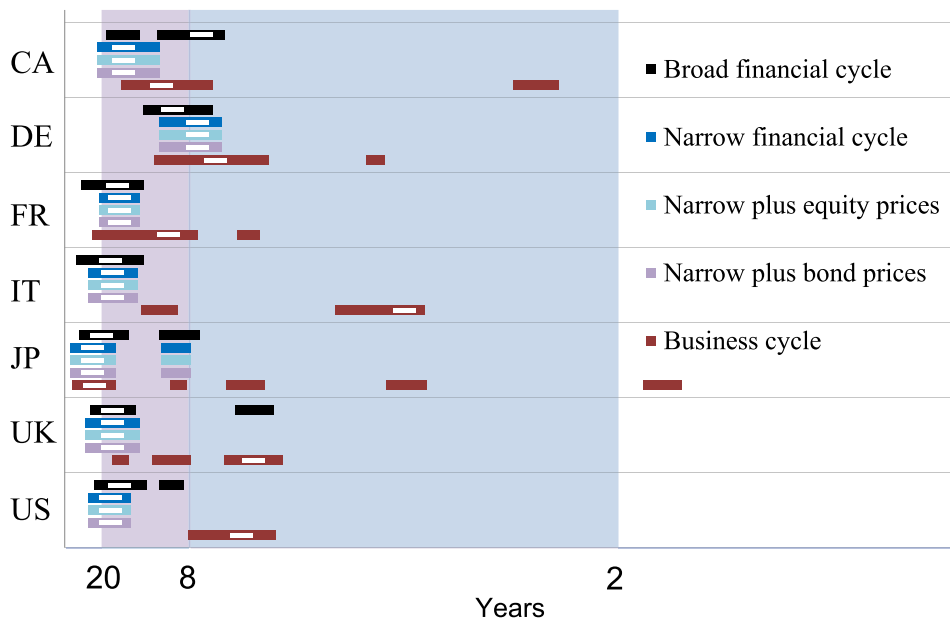


Fig. 3. Highest density chart of power cohesion. *Notes:* This graph depicts the 25% highest density region of power cohesion (see Hyndman, 1996). White bars denote the peak of power cohesion. Broad refers to the inclusion of all indicators, i.e. credit, house, equity, and bond prices, whereas narrow is defined by house prices and credit only. The x-axis measures the frequencies of cycles from 1.25 to 50 years. The blue area depicts business cycle frequencies, i.e. cycles with durations of 2–8 years, and the purple area (8–20 years) marks frequencies most important for financial cycles. For figure to appear in color, please see web version of this article.

relatively low home ownership rates (see Huber and Schmidt, 2019). We elaborate on this contradiction in the next part of the paper, where we report precise numbers of financial and business cycle frequencies, including frequency windows and HP smoothing parameters, obtained from PCoh, which can be used, for instance, to obtain financial cycle estimates through filtering.

Most important cycle duration, a frequency window, and an implied HP smoothing parameter.

Table 1 shows the duration in years of the most important cycle – the peak of PCoh – for the narrow and broad measure of the financial cycle, as well as the business cycle. In addition, we report a frequency window spanning that peak, defined by a maximum and minimum cycle length in years, that could be used, for instance, to inform band pass filters. Furthermore, we indicate in brackets implied HP smoothing parameters corresponding to the peak and the maximum cycle length respectively.⁹

Each frequency window is defined to capture 67% of the densest area around the peak of PCoh in the range of 1.25 and 50 years.¹⁰ The percentage is chosen on two grounds. First, considering 2/3 of the area can be argued to be agnostic, closely reflecting the smallest area that researchers are commonly interested in around a point estimate, in reference to the one-standard-deviation area around the mean of a normal distribution. Second, and more importantly, this specification yields a frequency window of the US narrow financial cycle that is commonly chosen in empirical studies identifying financial (growth) cycles. For instance, Drehmann et al. (2012) choose a window of 8 to 30 years, being close to our determined frequency window of 7.1 to 28.1 years in the case of the US narrow financial cycle. Furthermore, by construction, a maximum frequency of around 30 years corresponds to the cut-off frequency of a HP filter with a smoothing parameter of 400,000, as specified in Basel III for extracting the credit-to-GDP gap. Anchoring our methodology of determining a frequency window to recover the common US financial cycle filtering specification allows us to emphasise differences in financial cycles across countries, but also with respect to business cycles.

Across countries the narrow financial cycle lasts 15.6 years on average. The longest cycle is found in Japan (23.0 years), the shortest in Germany (7.4 years), as indicated in the previous parts. The average frequency window is 41 years to 4.9 years, with heterogeneity across countries. For instance, the minimum frequency ranges from 3.2 to 7.1 years, or the HP smoothing parameter for the maximum frequency ranges from 200,000 to 3,100,000, the upper limit allowed in this study that corresponds to a cycle length of 50 years.

⁹ More specifically, for this mapping we interpret each cycle length as the cut-off frequency of the HP filter, meaning around which the filter splits, for instance, a difference stationary time series in large samples, into cycle and trend (see Schüler, 2018).

¹⁰ This range is derived from commonly used specifications in the literature and economic theory. For instance, the widely used specification of the turning point algorithm requires a cycle to be at least 1.25 years in length. Furthermore, 50 years is the maximum length of cycles that can be derived from the phenomenon of Kondratiev waves. It is also assumed in Comin and Gertler (2006), amongst other studies. See Appendix A.4 for more details.

Table 1

Financial and business cycle frequencies, frequency windows, and implied HP smoothing parameters.

Country	Narrow financial cycle			Broad financial cycle			Business cycle		
	max.	peak	min.	max.	peak	min.	max.	peak	min.
Canada	50.0 ($3.1 \cdot 10^6$)	14.1 (19673)	3.4 –	50.0 ($3.1 \cdot 10^6$)	7.2 (1325)	2.7 –	15.4 (27786)	9.6 (4234)	2.1 –
Germany	50.0 ($3.1 \cdot 10^6$)	7.4 (1484)	3.2 –	50.0 ($3.1 \cdot 10^6$)	8.8 (2959)	2.9 –	45.6 ($2.1 \cdot 10^6$)	6.6 (960)	2.4 –
France	36.0 ($8.2 \cdot 10^5$)	15.0 (24682)	6.5 –	50.0 ($3.1 \cdot 10^6$)	15.4 (27786)	4.2 –	22.0 ($1.2 \cdot 10^5$)	9.1 (3402)	2.7 –
Italy	50.0 ($3.1 \cdot 10^6$)	16.4 (35608)	3.9 –	50.0 ($3.1 \cdot 10^6$)	16.9 (40552)	3.4 –	33.6 ($6.3 \cdot 10^5$)	3.2 (51)	2.8 –
Japan	50.0 ($3.1 \cdot 10^6$)	23.0 ($1.4 \cdot 10^5$)	5.2 –	50.0 ($3.1 \cdot 10^6$)	19.5 (71505)	3.9 –	50.0 ($3.1 \cdot 10^6$)	22.0 ($1.2 \cdot 10^5$)	2.1 –
UK	25.3 ($2.0 \cdot 10^5$)	16.4 (35608)	5.0 –	45.6 ($2.1 \cdot 10^6$)	16.4 (35608)	4.3 –	29.7 ($3.8 \cdot 10^5$)	5.4 (435)	3.1 –
US	28.1 ($3.1 \cdot 10^5$)	16.9 (40552)	7.1 –	33.6 ($6.3 \cdot 10^5$)	15.0 (24682)	4.9 –	15.4 (27786)	5.7 (541)	2.6 –
Avg.	41.3	15.6	4.9	47.0	14.2	3.8	30.2	8.8	2.6

Notes: The table shows the peak, maximum (max.), and minimum (min.) cycle length in years. The frequency range, defined by the maximum and minimum, captures 67% of the densest area around the peak. Figures in brackets depict the implied smoothing parameters for the HP filter, i.e. the cut-off frequency of the HP filter, about where trend and cycle of a difference stationary series are separated (see Schüler, 2018). Broad refers to the inclusion of all indicators, i.e. credit, house, equity, and bond prices, whereas narrow is defined by house prices and credit only. Avg. denotes the average.

With respect to the broad financial cycle, the area around the peak is generally less dense, so that on average the frequency window covers longer as well as shorter cycles, on average 47.0 to 3.8 years. The average cycle length is 14.2 years, and thus slightly shorter than the figure of the narrow financial cycle. While the peaks for the narrow and broad financial cycle can be argued to be similar for most countries, Canada represents an exception, with the average cycle length shrinking from 14.1 (narrow) to 7.2 years. As indicated by Fig. 3, this is due to an important common variation of equity and bond prices.

Calibrating our methodology to reflect common filtering specifications of the US financial cycle leads to the outcome that for business cycles the minimum frequency window is actually the one used in many studies, i.e. around 2.5 years. However, its maximum frequency is on average 30.2 years, rather than the commonly used parameter of eight years, pointing towards a greater importance of longer duration cycles in the context of business cycles, too. This supports the view of medium-term business cycles (Comin and Gertler, 2006; A'Hearn and Woitek, 2001). For the US, cycles up to 15.2 years should be considered, corresponding to an HP smoothing parameter of 27,786, which is in line with recent discussions on revising estimates of potential output by including longer-term fluctuations (see Phillips and Jin, 2015; Coibion et al., 2018). Closer to the commonly used HP smoothing parameter of 1600 is the parameter corresponding to the most important cycle length, which is 541. Finally, the average business cycle length across G-7 countries is 8.8 years, which, however, is heavily skewed by the very long business cycle in Japan (22 years). Excluding Japan, the average business cycle length falls to 6.6 years.

What may explain differences in financial cycles across countries? To arrive at an answer, Fig. 4a relates the home ownership rate to the duration of the narrow financial cycle.¹¹ We choose the narrow financial cycle, as Rünstler and Vlekke (2018) document a positive relation between the duration of credit and house price cycles with the home ownership rate.¹² As speculated above, Fig. 4a does not particularly favor this explanation, which is reflected, for instance, in an R^2 of 0.13. Rather, we find relatively strong evidence ($R^2 = 0.66$) for the view that higher home ownership rates relate to larger differences in financial and business cycle duration (Fig. 4b).

The case of Japan possibly impedes the relation uncovered in Rünstler and Vlekke (2018), who do not analyse this country. Furthermore, one could argue that the cases of Germany and Japan drive the elevated R^2 obtained for Fig. 4b, as these are the countries with almost no difference between financial and business cycles as well as the lowest home ownership rates. Thus, as robustness, we re-do both exercises excluding Germany and Japan. Still, our explanation is favored over the one provided by Rünstler and Vlekke (2018). For Fig. 4a, the R^2 reduces to 0.03 and for Fig. 4b to 0.25, which still ranks above the R^2 of Fig. 4a using all G-7 countries. We also find this compelling, as high ownership rates imply that a larger fraction of the population is sensitive to shocks to real estate, arguably leading to a stronger disconnect between financial and business cycles (see Hiebert et al., 2018).

¹¹ We consider the home ownership rates from Huber and Schmidt (2019), collected by PEW Research Center from national sources. For the US, we use the figures provided by the US Census Bureau for 2011, reflecting the most common reference year in Huber and Schmidt (2019).

¹² The authors do not consider cycles from equity and bond prices.

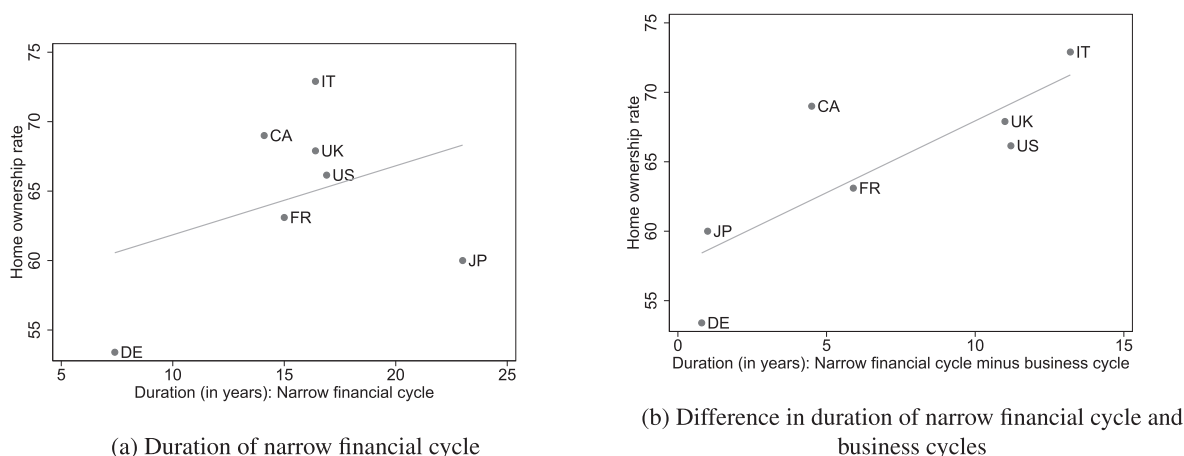


Fig. 4. Home ownership rates and financial cycle frequencies. *Notes:* Solid lines stem from a linear regression of the home ownership rate on the other indicated variable. Home ownership rates are taken from Huber and Schmidt (2019) that are collected by PEW Research Center from national sources. For the US, home ownership rates are from US Census Bureau in 2011, reflecting the most common reference year in Huber and Schmidt (2019).

4. Financial cycles across time

Having established that durations of financial cycles vary considerably across countries, in this section, we proceed by presenting a methodology for constructing a composite financial cycle indicator that relaxes the assumption of similar cycle duration across countries that is implicit, for instance, in the construction of the Basel III credit-to-GDP gap.

To this end, we transfer the concept of co-movement of indicators from the frequency domain to the time domain, albeit with one difference: When combining indicators into an index in the time domain, lead-lag relations matter. As we would like to advance a methodology for constructing a real-time indicator, we refrain from rigidly modelling lead-lag relations between indicators (as they differ across countries – see Appendix A.2 – and most likely also over time). Thus, we decide to aggregate indicators contemporaneously.

Based on our proposed composite financial cycle indicators, we analyse financial cycle synchronisation across countries and seek to find out whether the indicators, which track common expansions and contractions in credit and asset prices, can be used both as a coincident and as an early warning indicator of systemic banking crises, possibly improving on the Basel III credit-to-GDP gap. First, however, we lay out our methodology for constructing a composite indicator.

4.1. Methodology: Constructing a composite financial cycle indicator

Time-varying aggregation of standardised indicators.

The collation of the filtered growth rates (that allow for all cycle durations up to 50 years) of indicators into a composite financial cycle index requires two steps. First, we standardise variables given different measurement units, but also in light of dissimilar variances and equilibrium growth rates. Second, we aggregate these standardised indicators into composite financial and business cycle series.

In keeping with the nomenclature of the previous sections, we distinguish between a narrow composite financial cycle (consisting of credit and house prices) and a broad measure (combining credit, house, equity, and bond prices). For these representations, as well as analogous measures for the business cycle, we recursively construct time-varying aggregation weights that exploit changing correlation structures between variables. Due to the recursivity of our approach, the method is especially suited for constructing real-time indicators required for policy making. In doing so, we are closely related to the methodology followed by Holló et al. (2012) for constructing an indicator of systemic financial stress.¹³ Our approach is specifically suited for modelling cyclical fluctuations, while Holló et al. (2012) capture spikes in financial stress.

More formally, we transform respective underlying cycle indicators using their empirical cumulative distribution function (ecdf), whereby each value of a single indicator is mapped into a unit free ordinal scale in $(0,1]$. Let $(x_{i,[1]}, x_{i,[2]}, \dots, x_{i,[T]})$ denote the ordered sample of variable $x_{i,t}$. The transformed indicators are derived by

¹³ One alternative approach for aggregating common information contained in several variables is through principal component analysis. This approach has, however, at least two drawbacks. First, cross-country comparability issues arise due to a possible different selection of indicators across countries that explain most important contemporaneous movements. Second, real-time updating is an issue in that new data points might alter the selection of indicators most relevant for common fluctuations. Indeed, a principal component analysis is static by definition and thus does not allow for capturing changing trends in the interdependence between financial indicators.

$$y_{i,t} = F_{i,T}(x_{i,t}) = \begin{cases} \frac{r}{T} & \text{for } x_{i,[r]} \leq x_{i,t} < x_{i,[r+1]}, \quad r = 1, 2, \dots, T-1 \\ 1 & \text{for } x_{i,t} \geq x_{i,[T]}, \end{cases} \quad (3)$$

where $F_{i,T}$ denotes the empirical cdf.

These standardised indicators are then taken to construct synthetic cycles by linearly combining these using time-varying weights that exploit *positive* time-varying correlations between indicator pairs. To this end, let C_t reflect the $M \times M$ matrix of time-varying cross-correlations. While the diagonal elements of C_t are one, the off-diagonal elements, $c_{ij,t}$ with $i \neq j$, are given by

$$c_{ij,t} = \begin{cases} \rho_{ij,t} = \sigma_{ij,t} / (\sqrt{\sigma_{ii,t} \sigma_{jj,t}}) & \text{if } \sigma_{ij,t} \geq 0 \\ 0 & \text{if } \sigma_{ij,t} < 0 \end{cases}, \quad (4)$$

where $\sigma_{ij,t} = \lambda \sigma_{ij,t-1} + (1 - \lambda)(y_{i,t} - 0.5)(y_{j,t} - 0.5)$, $y_{i,t}$ is centered using the median and λ represents the decay factor that we set to 0.89.¹⁴

To define ζ_t , i.e. the composite cycle indicator, let $Y_t = (y_{1,t}, \dots, y_{M,t})$ and ι be a vector of ones of dimension $M \times 1$. Then,

$$\zeta_t = \frac{1}{\iota' C_t \iota} \cdot \iota' C_t Y_t, \quad (5)$$

which implies, for instance, for expositional purposes in the three-variable case ($M = 3$) and exploiting the fact that $c_{ij,t} = c_{ji,t}$,

$$\zeta_t = \frac{(1 + c_{12,t} + c_{13,t})y_{1,t} + (1 + c_{12,t} + c_{23,t})y_{2,t} + (1 + c_{13,t} + c_{23,t})y_{3,t}}{3 + 2c_{12,t} + 2c_{13,t} + 2c_{23,t}}. \quad (6)$$

Due to the aggregation method, it holds that $\zeta_t \in (0, 1]$, like the underlying transformed indicators.¹⁵ Real-time updating of ζ_t (see Section 4.4.1) is carried out by applying Eq. (3) recursively over expanding samples, i.e. “old” observations are fixed and new ones are added to the existing distribution.

Restricting correlations as in Eq. (4) means that our aggregation scheme emphasises positively related or identical movements. While all series enter the final index by construction (see, for instance, Eq. (6)), it implies that the series, whose movements are positively related, receive higher weights. To illustrate, consider the case of three indicators: if two of the series remain above (or below) their historical medians and the third series is located below (or above) its historical median for an extended period of time, using our method the final signal tends to reflect the location of the first two series, since the third series is negatively related and negative correlations are assumed to be zero. As a consequence, the overall aggregation weight of the third series is lower relative to the first two – but still larger than zero. Employing unrestricted correlations, negative and positive correlations would cancel each other out and the aggregation would be similar to a simple average. In this case, the location of the financial cycle would not emphasise the build-up or correction of imbalances among a subset of indicators. This, however, is important, as evidenced by common fluctuations especially in credit and house prices prior to the global financial crisis.

Filtering: full-sample and real-time.

It is important to note that the combination of quarterly standardised growth rates to form a composite indicator already acts as a filtering device, since idiosyncratic movements are dampened and common fluctuations emphasised (also beyond the time-varying aggregation scheme).

Still, for a policy indicator the signal obtained from the combination of quarterly standardised growth rates can be argued to be too volatile, as quarterly growth rates allow for all sorts of cycle durations. Thus, we apply one of two different filtering procedures to smooth the composite indicators, depending on its use.

The first procedure is suited for full-sample analyses, the second for real-time analyses. In the first case, we filter the composite indices using the country-specific frequency bands based on PCoh obtained on the full sample (see Table 1) as an input to the band pass filter proposed by Christiano and Fitzgerald (2003). In the second case, we exploit a weighted moving average of six quarters. We employ declining weights that are based on a one-sided Bartlett window that gives the highest weight to the last observation at time t , arguably important for real-time policy analysis. Weights are $1 (\zeta_t)$, $5/6 (\zeta_{t-1})$, $4/6 (\zeta_{t-2})$, $3/6 (\zeta_{t-3})$, $2/6 (\zeta_{t-4})$, and $1/6 (\zeta_{t-5})$ scaled by their sum to keep the scale $(0,1]$. For constructing real-time estimates, we prefer this method, as the Christiano and Fitzgerald (2003) band pass filter suffers from an endpoint bias, which is especially problematic in real-time applications.

¹⁴ This is slightly smaller than in Holló et al. (2012), who use 0.93. We employ a smaller decay factor since we have quarterly (instead of daily) data with, thus, fewer observations. Using 0.89, starting conditions become negligible at a faster rate. Further, covariances are initialised using the first eight quarters of observations for each variable.

¹⁵ For expositional purposes, we re-map ζ_t into $(0,1]$ using Eq. 3. As the case of all four indicators being close to their historical maxima or minima is unlikely, the amplitude of the not yet re-mapped indicator ζ_t is dampened towards the median.

4.2. Composite cycles

Fig. 5 depicts the composite financial and business cycle indices for the US in deviations from historical median growth. As indicated, the mere aggregation produces a strong cyclical impetus, from which we further eliminate high frequency fluctuations by means of filtering (see Fig. 5a). The right-hand panel complements this information, showing both the narrow and broad composite financial cycle as well the business cycle.

A divergence of US financial and business cycles during upswing phases (growth rates above 0.5) seems to precede systemic banking crises. Specifically, Laeven and Valencia (2012, 2018) define two US systemic banking crises for our sample period, i.e. the savings and loan crisis in 1988 (borderline case) and the global financial crisis in 2007. Before both crises a divergence of the financial (both broad and narrow) and the business cycle is visible. Or put differently, the upswing phases of the financial cycles preceding crises are especially long.

Furthermore, recovery periods are longer for both financial cycles than for the business cycle. In this context, Hiebert et al. (2018) provide evidence of this being the most important dimension of divergence between financial and business cycles.

Nonetheless, there is an apparent co-movement between the two cycles, which is in line with both our analysis of statistical properties (Appendix A.2) and the literature, emphasising a medium-term link between financial and business cycles (see Rünstler and Vlekke, 2018) or the importance of financial cycle indicators' downturns in shaping recessionary phases (see Claessens et al., 2012).

These results are broadly consistent with the findings for the remaining country composite cycles (see Appendix A.6). Interestingly, the composite financial cycles of Germany and Japan follow very distinct movements around the global financial crisis. For Germany this means that, since the crisis was imported, the index does not indicate strong imbalances prior to the crisis, as would be indicated by high deviations from median growth. However, we find evidence of an expansion after the outbreak of the global financial crisis. This also holds true for Japan, for which country case no (global) financial crisis is reported by Laeven and Valencia (2012, 2018). Here, both financial cycles similarly expand over that period, while its business cycle collapses. In the case of the UK, financial and business cycles are quasi identical until a peak located around 1990, from which diverging cycles emerge.

4.3. Synchronisation of financial cycles across countries

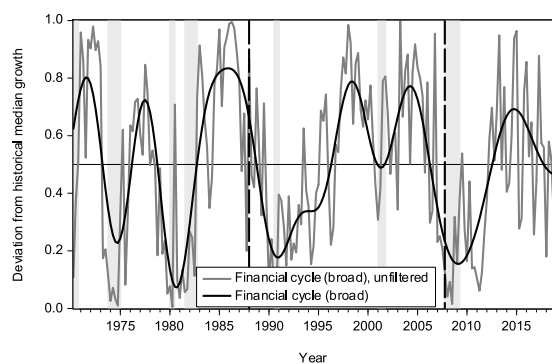
Next we consider how a G-7 financial cycle (narrow and broad) and a G-7 business cycle correlate with their respective country cycles. To this end, we construct the G-7 cycles as the first dynamic principal component of the respective country cycles (see Fig. 6 (a)). In such a time series context, dynamic principal components are preferred over static principal components as they take autocorrelations into account. Specifically, we obtain the dynamic principal components using the two-sided factor decomposition as proposed by Forni et al. (2000).

In the case of the narrow financial cycle, a first principal component summarises 51% of the common variation; in the case of the broad financial cycle 58%, and in the case of the business cycle 57%.¹⁶ The narrow and broad G-7 financial cycles are strongly correlated, with a correlation coefficient of 0.81. The G-7 business cycle relates more weakly to the financial cycles, with a correlation coefficient of 0.48 with the broad indicator and of only 0.21 with the narrow indicator. This is in line with Breitung and Eickmeier (2014), who find evidence of a global factor affecting both business and financial cycles, while noting that there is still an important financial factor independent of the macro factors affecting financial variables.

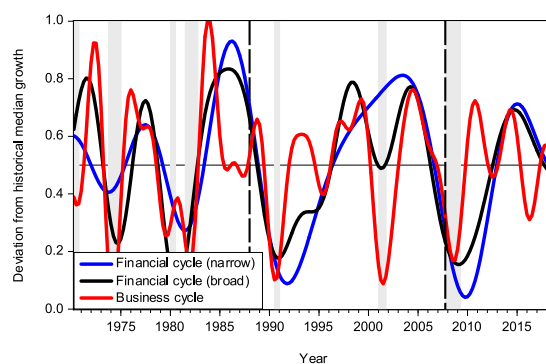
Similar to the US financial cycle, we observe phases of coupling and de-coupling of financial and business cycles. For instance, prior to the global financial crisis, financial cycles are in their upswing phase (above 0.5) and decouple from business cycles that are in a downswing phase. As most of the countries have been jointly affected by the global financial crisis, it is visible in the G-7 financial cycle (see dashed vertical lines, indicating systemic banking crises of the G-7 countries as defined by Laeven and Valencia (2012, 2018)).

In the right-hand panel of Fig. 6 we correlate the G-7 cycles with the respective country cycles, i.e. for instance the G-7 financial cycle (broad) with the financial cycle (broad) of France. At least two results are worth emphasising. First, while G-7 financial cycles (narrow and broad) relate heterogeneously to the country financial cycles, business cycles are rather homogeneously related across G-7 countries. Specifically, the G-7 financial cycle (broad) correlates strongly with the country financial cycles (broad) of France (0.89), Italy (0.80), and Canada (0.78), to a lesser extent with those of the UK (0.72) and the US (0.64). It correlates even less strongly with the country financial cycles (broad) of Germany and Japan (0.48 and 0.44). Results are more heterogeneous when considering the narrow measures. Correlated with the G-7 financial cycle (narrow) are the country financial cycles (narrow) of France (0.94), Italy (0.83), Canada (0.77), and the UK (0.69). Less correlated are the country financial cycles (narrow) of the US (0.47), Japan (0.27), and Germany (0.18). The weak correlation of Germany and Japan to a common component dovetails with the findings of the individual country financial cycles noted in Section 4.2 or their distinct frequencies (see Section 3). This might be partly related to their distinct developments in both the housing and credit markets in recent years. These are the only countries where house prices have been declining alongside stagnating

¹⁶ The second principal components capture only 19%, 16%, and 16% respectively. Note that the arguably low fractions of variance explained by the first principal components are likely explained by the fact that we draw the principal components from the filtered composite indicators, where shorter-term fluctuations are completely removed. That is, shorter-term fluctuations may explain a large part of co-movement across countries, for instance, as is the case with equity prices.

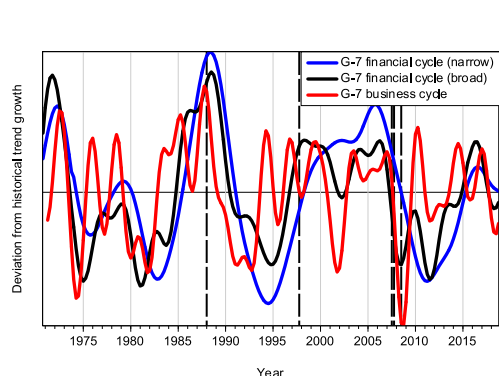


(a) Filtered and unfiltered financial cycle, broad

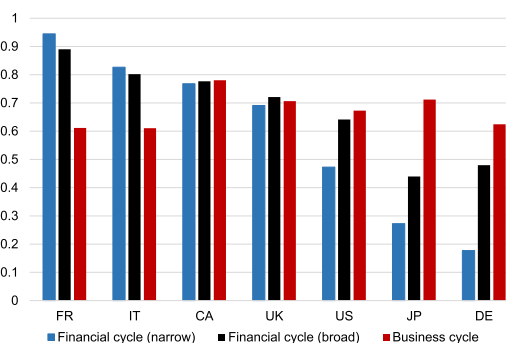


(b) Filtered composite cycle indices

Fig. 5. US composite cycles. *Notes:* This panel shows the US composite financial and business cycles in standardised growth rates, where 0.5 denotes the historical median; 0 is the smallest and 1 the largest growth rate observed in a country's history. The left-hand graph depicts the broad composite financial cycle (filtered and unfiltered) combining information from credit, house prices, equity prices and bond prices. The right-hand graph shows the filtered financial (narrow and broad) and business cycles. Filtering is done using the [Christiano and Fitzgerald \(2003\)](#) band pass filter employing country-specific frequency windows as described in [Table 1](#). Aggregation of standardised indicators is done using linear averaging for the narrow financial cycle and using time-varying weights that emphasise alike movements between indicators for the broad financial cycle. See [Section 4.1](#) for further details. The grey area indicates NBER recession dates. Black vertical dashed lines indicate the onset of systemic banking crises as defined by [Laeven and Valencia \(2012, 2018\)](#).



(a) G-7 financial (narrow and broad) and business cycles



(b) Correlation of respective G-7 with respective country cycles

Fig. 6. G-7 cycles and correlation with respective country cycles. *Notes:* The left-hand panel depicts the first dynamic principal component of the countries' financial and business cycles. Black dashed vertical lines indicate the onset of systemic banking crises as defined by [Laeven and Valencia \(2012, 2018\)](#) of the set of G-7 countries. The right-hand panel shows the correlation of country financial (narrow and broad) and business cycles with the respective first principal components. Countries are ordered by highest correlation of financial cycles (broad) with the G-7 financial cycle (broad) principal component. For figures to appear in color, please see web version of this article.

or declining residential investment at least until 2009 (see [André, 2010](#); [Knoll et al., 2017](#)). In line with our argument that joint movements in credit and asset prices matter, it is also Germany and Japan that do not experience a credit boom during that period (see [Hume and Sentance, 2009](#)). All of this chimes well with our evidence reported in the previous section: Germany and Japan are marked by the lowest home ownership rates, and are thus possibly not as strongly affected by global shocks to real estate or credit markets. In contrast, business cycles relate homogeneously where correlation spans between 0.78 and 0.61. The strongest link to the common business cycle component is shared by Canada and the weakest by France and Italy (both 0.61), which, however, is still elevated compared with the results of financial cycles.

Second, financial assets (equity and bond prices) are an important link for financial cycles across countries. In general, total interlinkage is stronger among the G-7 countries when considering a broader class of assets in line with the studies of [Rey \(2015\)](#) and [Breitung and Eickmeier \(2014\)](#). Explanations entail, for example, that credit and house prices are more of a domestic component. For instance, houses are not tradable and are subject to strong country-specific regulations also with respect to financing.

In sum, we find that financial cycle synchronisation across G-7 countries is heterogeneous, relative to the link of G-7 business cycles. Germany and Japan are the least linked. In line with previous studies, we find that financial asset prices account for strong international linkages.

4.4. Coincident and early warning indicators of systemic banking crises

To validate the composite financial cycle indicators, we conduct a signalling exercise testing their performance both as coincident as well as early warning indicators of systemic banking crises. We compare their performance to the underlying indicators and to the credit-to-GDP gap. In the following we lay out the methodology for conducting our signalling exercise and subsequently present the results.

4.4.1. Methodology: Pooled logit

The signalling approach is based on the predicted probabilities of a logit model estimated on a pooled data set to explain the start of systemic banking crises as defined by [Laeven and Valencia \(2012, 2018\)](#) (seven in the sample) as well as two vulnerability periods prior to the onset of crises.¹⁷

Specifically, we create one dummy variable indicating the onset of a crisis, one indicating one to four quarters before the onset of a crisis, and one indicating five to twelve quarters before the onset of crisis; and zero otherwise. As several studies – in reference to a so-called “post-crisis bias” – remove the start of crisis as well as six quarters thereafter from their vulnerability dummy variables (e.g. for a five to twelve quarters early warning period in [Anundson et al. \(2016\)](#)), we report results of such specification in a robustness check in [Appendix A.8](#).¹⁸

We assess the signalling properties of several indicators – some separately, others jointly – using both a pseudo-out-of-sample and an in-sample setup. We put more emphasis on the pseudo real-time analysis since it mirrors more closely the decisions faced by policy makers. As separate indicators we use the financial and business cycle indicators, the credit-to-GDP gap, and credit growth. We consider credit growth and house price growth jointly, as well as the latter indicators plus equity price growth and bond price growth, mirroring the information set contained in our narrow and broad composite financial cycles.

In the *pseudo out-of-sample* exercise we estimate the logit model on data from 1980Q1 until 1999Q4 – using the period prior to 1980Q1 to initialise the standardisation of indicators (see Eq. (3)) – and conduct the out-of-sample exercise on an expanding sample for the period thereafter. For each sample $(1, \dots, t)$, we forecast the one-quarter ahead predicted probabilities $(t + 1)$ for individual countries and check separately whether they breach an optimal in-sample signalling threshold, calling a crisis event. The in-sample signalling threshold value is optimally determined by minimising a loss function that gives equal importance to avoiding Type I (TI, missing crisis or vulnerability period) and Type II errors (TII, false alarms) errors, i.e. $L = TI + TII$. In this manner, we collect the true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) on an expanding sample. Using these, we derive the out-of-sample Type I and II errors, the relative usefulness (U^r), and the noise-to-signal ratio (NtS). The TI error can be defined as the false negative rate, i.e. $FN/(FN + TP)$. Analogously, the TII error reflects the false positive rate, i.e. $FP/(FP + TN)$. Relative usefulness is defined as $(0.5 - L)/0.5$ and measures the relative gain in discarding the indicator(s) (see [Alessi and Detken \(2011\)](#)). The noise-to-signal ratio is captured by $TII/(1 - TI)$, relating two ratios: the ratio of bad signals to all quarters without crisis (noise) to the ratio of good signals to all quarters with crisis (signal) (see [Kaminsky et al., 1998](#)).

In addition, we calculate the area under the curve (AUC) considering the out-of-sample predicted probabilities. The AUC relates the true positive rate ($TP/(FP + TP)$) to the false positive rate for every threshold value on the real line. Thus, the AUC has the advantage over the above measures of summarising an indicator's performance for all possible threshold values. It is not sensitive to ad hoc preferences given to avoiding TI and TII errors. An indicator based on a coin toss would lead in large samples to an AUC of 0.5. A perfect indicator, on the other hand, would without error discriminate between crisis and non-crisis periods, leading to an AUC of 1. As we are interested in comparing the signalling performance of our indicators to that of the credit-to-GDP gap, we test whether the AUC of an indicator is significantly different to the AUC of the credit-to-GDP gap. For this purpose, we employ the test proposed by [DeLong et al. \(1988\)](#), which accounts for possible correlation between two curves.

To be more precise, for the pseudo out-of-sample exercise let $\tilde{\pi}_{lt} = P(\tilde{y}_{lt} = 1 | \mathcal{F}_{t-1})$, where $\tilde{y}_{lt}$ refers to a crisis dummy for country l at time period t . $\mathcal{F}_{t-1}$ is the information set containing information up to time period $t - 1$. We predict the one-quarter ahead probabilities using the lagged relationship

$$\text{logit}(\tilde{\pi}_{lt}) = \phi_* + \sum_{k=1}^K \tilde{X}_{l,t-k} \phi_k, \quad (7)$$

where $\text{logit}(\tilde{\pi}_{lt}) = \ln(\tilde{\pi}_{lt}/(1 - \tilde{\pi}_{lt}))$ is the log of the odds ratio, ϕ_* is the intercept, ϕ_k a vector of coefficients, and $\tilde{X}_{l,t-k}$ is a $1 \times M$ vector with M predictors at time period $t - k$. We allow for a maximum number of lags of 6. The lags are allowed to vary across the sets of indicators (not across time) and are determined via maximising the AUC.

¹⁷ We use the quarterly starting dates of systemic banking crises. Similar to Section 4.2, we include the US 1988 crisis, which is described as a borderline case by [Laeven and Valencia](#) and [Valencia \(2012, 2018\)](#) and use the first quarter as its starting date. In sum, our sample includes two US crises and one crisis in Germany, France, Italy, Japan, and the UK respectively; there were none in Canada.

¹⁸ We actually believe that a financial cycle indicator should clearly signal the onset of financial crises, analogously to a business cycle indicator clearly identifying recessionary periods. Therefore, we argue that it is important to include the period after the onset of crisis as zeros in the dataset.

In the *in-sample* exercise, we discard the publication lag as all data are assumed to be available, i.e. we consider the predictors contemporaneously. We still use lagged values of indicators, now with $K \in \{0, 1, 2, 3, 4, 5\}$. In this exercise, we only summarise the signalling performance via the AUC. Again, we test whether an indicator's AUC is significantly different to the AUC of the credit-to-GDP gap.

For both exercises we employ the real-time estimates of our composite financial cycles (see Section 4.1).¹⁹ For comparative purposes, the real-time cycles of the US are depicted in Fig. 7. The remaining countries are shown in Appendix A.7. Note that while these are more volatile than the Christiano and Fitzgerald (2003) band pass filtered cycles, the changes in cyclical phases are still clearly visible.

4.4.2. Results

The signalling results reinforce the case for aggregation, suggesting that combining information helps to improve the capacity of financial variables to predict the start and vulnerability periods prior to systemic banking crises, especially in real time and especially considering the broad set of asset prices. That is, we report that, among the indicators, the broad financial cycle reflects the best ranking one. It consistently ranks first and significantly outperforms the credit-to-GDP; with the exception of the *in-sample* exercise at the very long horizon (five to twelve quarters ahead).

Panel A of Table 2 offers the results for the signalling exercise for the start of the crisis. The left part of the table, "One quarter out-of-sample" suggests that the broad financial cycle index is the best coincident indicator in real time, while the narrow financial cycle ranks second. The broad measure allows us to detect 3 (TP) out of 5 (TP + FN) crisis starting dates correctly. These are the onsets of the global financial crises in France, Italy, and the US. It misses the onsets in Germany and the UK. Note that – following Laeven and Valencia (2012, 2018) – Canada and Japan did not experience a crisis during our out-of-sample period, i.e. past 1999Q4. Given the lowest number of false positives, the relative usefulness of the broad measure is highest (0.47) and its noise-to-signal ratio lowest (0.21; lowest value indicating best performance). The AUC is 0.81, being significantly different to that of the credit-to-GDP gap (0.38) at the 1% significance level. The credit-to-GDP gap fails to indicate the onset of the global financial crises in Germany, the UK, and the US. Furthermore, it has a larger share of false positives than the broad financial cycle. All other sets of variables have an AUC that does not significantly differ to the AUC of the credit-to-GDP gap, except for the business cycles indicator at the 10% level. Its AUC (0.53), however, is of similar magnitude than the AUC of a coin toss in large samples.²⁰

In-sample, the broad financial cycle measure, again, receives the highest AUC (0.90), which significantly differs to the AUC of the credit-to-GDP gap (0.71) at the 10% significance level. Next, credit with the entire set of asset prices as well as credit with house prices are statistically different to the credit-to-GDP gap with an AUC of 0.89 and 0.82 respectively. This stresses the role of asset prices in predicting crises, but also the need for aggregation for good performance in real time. Lastly, for the remaining indicators we do not find a significant difference to the signalling performance of the credit-to-GDP gap.

Panel B of the same table shows the results of the early warning exercise considering the one to four-quarters-ahead window. Out-of-sample, the broad financial cycle indicator has the highest AUC (0.72), the second highest U^r (0.39), and the third lowest NtS ratio (0.40). Again, the AUC is significantly different to that of the credit-to-GDP gap (0.53) at the 1% significance level. For this vulnerability horizon, the narrow financial cycle reaches an AUC (0.67), which is quite close to the broad measure and is also significantly different to the AUC of the credit-to-GDP gap at the 1% level. Both financial cycles (narrow and broad) predict at least one quarter of a country's vulnerability period correctly, in contrast to the credit-to-GDP gap that does not signal any of the four vulnerability quarters in the case of Germany and France. The underlying indicators, as well, provide a signal for all crisis-hit-countries, however calling fewer true positives.

In-sample, the broad financial cycle measure does – as for the start of a crisis – slightly better than when considering its underlying indicators (0.85 to 0.82). The AUCs of the composite and underlying indicators are significantly different to the AUC of the credit-to-GDP gap (0.60) at the 1% significance level. In this exercise, the narrow financial cycle ranks third with an AUC of 0.74 (significantly different at the 5% level and exactly the same AUC as credit with house prices).

Panel C reports results for the period five to twelve quarters ahead of a crisis. Out-of-sample, the broad measure ranks first considering the AUC (0.64) alongside credit with house prices. It ranks third looking at the NtS ratio (0.34), and fifth looking at U^r (0.20). In this exercise, the broad measure fails to signal any of the vulnerability quarters in Germany and the UK, while the credit-to-GDP gap does not signal any vulnerability quarter only in Germany. Still, the AUC of the broad measure is found to be significantly different to that of the credit-to-GDP gap (0.55) at the 1% level, because it has by far lower false positives than the credit-to-GDP gap; 50 versus 270. 50 actually reflects the lowest false positive rate, lower than credit with house prices, with a count of 60. The AUC of the narrow financial cycle ranks third (significantly different at the 1% level). This indicator only misses Germany.

¹⁹ As indicated in Section 2, for the real-time estimates we use the unfiltered, i.e. standard, growth rates of indicators to construct the composite cycle indicators.

²⁰ Interestingly, the AUC of the second-ranked indicator (0.57: narrow financial cycle) is not significantly different to the AUC of the credit-to-GDP gap, but the AUC of the third-ranked indicator (0.53: business cycle) is. This is because the curves resulting from the business cycle index and the credit-to-GDP gap are only weakly related, while the curves of the narrow financial cycle and the credit-to-GDP gap are more strongly related, which increases the confidence bounds for the latter test.

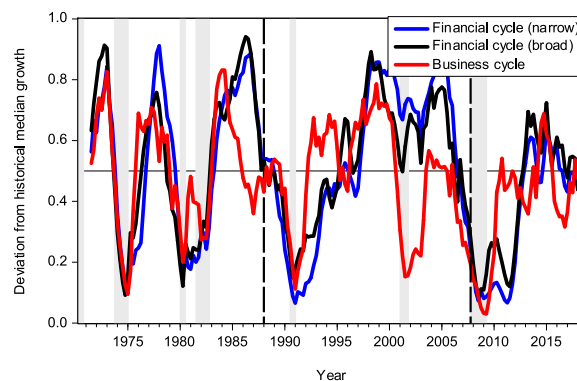


Fig. 7. US real time composite financial and business cycles. *Notes:* The real-time US composite financial and business cycles are measured in standardised growth rates, where 0.5 denotes the historical median; 0 is the smallest and 1 the largest growth rate observed in a country's history. Smoothing is done using a six-quarter one-sided moving average based on the weights of a Bartlett window. See Section 4.1 and 4.4.1 for further details. The grey area indicates NBER recession dates. Black vertical dashed lines indicate the onset of systemic banking crises as defined by Laeven and Valencia (2012, 2018). For figures to appear in color, please see web version of this article.

In-sample, the broad and narrow measures rank fourth with an AUC of 0.66, which are not significantly different to the AUC of the credit-to-GDP gap (0.62). The joint inclusion of credit and all asset prices performs best (0.79) and statistically outperforms the credit-to-GDP gap at the 1% significance level.

The results of the robustness exercise (Appendix A.8), for which we remove observations that fall on the start of a crisis as well as six quarters thereafter, paint a similar picture. Considering the one to four-quarters ahead window, as well as the five-to-twelve quarters ahead window, the broad and narrow financial cycles significantly outperform the credit-to-GDP gap in the out-of-sample exercise. Similar to the main analysis, in the in-sample exercise the composite financial cycle indicators only outperform the credit-to-GDP gap in the one-to-four-quarters ahead window.

Table 2

Signalling exercise: Coincident and early warning.

Indicators	One quarter out-of-sample											In-sample		
	Obs.	K	TP	FP	TN	FN	TI	TII	U ^r	NtS	AUC	Obs.	K	AUC
Panel A: At start of crisis														
Financial cycle (narrow)	532	4	2	79	448	3	0.60	0.15	0.25	0.37	0.57	1057	4	0.77
Financial cycle (broad)	532	4	3	67	460	2	0.40	0.13	0.47	0.21	0.81***	1057	5	0.90*
Business cycle	532	6	1	130	397	4	0.80	0.25	-0.05	1.23	0.53*	1057	5	0.82
Δcr	532	6	1	104	423	4	0.80	0.20	0.00	0.99	0.43	1057	5	0.72
Δcr & Δp_h	532	6	1	115	412	4	0.80	0.22	-0.02	1.09	0.41	1057	5	0.82*
Δcr , Δp_h , Δp_e , & Δp_b	532	2	0	130	397	5	1.00	0.25	-0.25	-	0.38	1057	5	0.89*
Credit-to-GDP gap	532	1	2	105	422	3	0.60	0.20	0.20	0.50	0.38	1057	2	0.71
Panel B: 1–4 quarters before crisis														
Financial cycle (narrow)	532	5	14	256	256	6	0.30	0.50	0.20	0.71	0.67***	1057	5	0.74**
Financial cycle (broad)	532	3	15	183	329	5	0.25	0.36	0.39	0.48	0.72***	1057	5	0.85***
Business cycle	532	3	9	247	265	11	0.55	0.48	-0.03	1.07	0.57	1057	5	0.66
Δcr	532	6	12	109	403	8	0.40	0.21	0.39	0.35	0.62**	1057	5	0.73*
Δcr & Δp_h	532	6	13	124	388	7	0.35	0.24	0.41	0.37	0.56	1057	5	0.74**
Δcr , Δp_h , Δp_e , & Δp_b	532	4	10	196	316	10	0.50	0.38	0.12	0.77	0.48	1057	5	0.82***
Credit-to-GDP gap	532	1	8	78	434	12	0.60	0.15	0.25	0.38	0.51	1057	0	0.63
Panel C: 5–12 quarters before crisis														
Financial cycle (narrow)	532	2	19	136	356	21	0.53	0.28	0.20	0.58	0.63***	1057	5	0.66
Financial cycle (broad)	532	1	12	50	442	28	0.70	0.10	0.20	0.34	0.64***	1057	5	0.66
Business cycle	532	1	34	315	177	6	0.15	0.64	0.21	0.75	0.48***	1057	5	0.59
Δcr	532	2	22	71	421	18	0.45	0.14	0.41	0.26	0.61**	1057	5	0.67**
Δcr & Δp_h	532	2	23	60	432	17	0.43	0.12	0.45	0.21	0.64***	1057	5	0.71***
Δcr , Δp_h , Δp_e , & Δp_b	532	4	19	206	286	21	0.53	0.42	0.06	0.88	0.53	1057	5	0.79***
Credit-to-GDP gap	532	1	31	270	222	9	0.23	0.55	0.23	0.71	0.55	1057	5	0.62

Notes: Table shows results of the out-of- and in-sample exercise as described in Section 4.4.1. "Obs." refers to observations, "K" to the number of lags, "TP" to true positives, "FP" to false positives, "TN" to true negatives, "FN" to false negatives, "TI" to Type I error, "TII" to Type II error, "U^r" to relative usefulness, "NtS" to noise-to-signal ratio, and "AUC" to area under the curve. *, **, and *** indicate that the AUC of the respective indicator is significantly different to the AUC of the credit-to-GDP gap at the 10%, 5%, and 1% significance level, using the test proposed by DeLong et al. (1988) that accounts for correlation between two curves. "-" indicates that the statistic is not defined.

Overall, this exercise points to the merits of combining information for a real-time exercise at any of the considered signalling horizons, even though results are stronger for periods smaller than the five-two-twelve quarters horizon. Findings support the view that idiosyncratic movements in individual indicators are dampened in constructing the composite financial cycles and, thus, the signalling performance in real time increases over considering individual indicators. In-sample, individual indicators perform similarly, or better over the long horizon. Evidence that the broad financial cycle is the best performing indicator suggests that a larger set of asset prices than house prices is important for real-time financial crisis prediction. Arguably, this reflects the forward-looking nature of financial assets, as suggested by the descriptive offered in [Appendix A.2](#). Finally, the results support our evidence against the similar-duration assumption inherent in the credit-to-GDP gap. For Germany, where financial cycles deviate most strongly from the assumed duration in Basel III regulations, the credit-to-GDP gap fails to signal at any of the periods. This is in contrast to the proposed composite cycles as well their underlying indicators.

5. Conclusions

Accurate measurement of country financial cycles is critical to underpin countercyclical macroprudential policy mandates. In this study, we show that financial cycles differ considerably in length across countries using a novel empirical method called power cohesion. This is in contrast to the assumption of similar duration cycles inherent to commonly used measures, such as the Basel III credit-to-GDP gap designed to guide countercyclical capital buffers. Against this backdrop, we propose an empirical approach to construct composite financial cycles exploiting the common fluctuation of credit and asset prices. The method relaxes the similar-duration assumption across countries and across time. It is specifically suited for real-time analysis. We find that the constructed composite financial cycles significantly outperform the credit-to-GDP gap in predicting systemic banking crises.

Three main policy conclusions follow from our results. First, country-specific policies to act countercyclically within national financial systems are supported by heterogeneous financial cycles across countries. That said, a prominent role needs to be accorded to reciprocity, amid spillovers and common drivers, given a strong synchronisation of financial cycles evident for some country groupings. Second, policy separation – whereby macroprudential policies cover financial cycles and monetary policy business cycles – is called for the longer length and higher amplitude of financial cycles compared with business cycle counterparts. The case for macroprudential policy addressing the build-up of systemic risk at the national level to complement policies aimed at business cycle stabilisation is especially strong for euro area countries, in which business cycles link closely, in contrast to marked differences in financial cycle synchronisation. This is not to say that monetary policy effectiveness is independent of financial cycles; the credit channel of monetary policy is likely to be far more effective in high states of the financial cycle than in low states. In high states of the financial cycle, firms and households, due to increased leverage and larger-sized balance sheets, become much more sensitive to variations in both asset prices and credit supply. In contrast, in low states of the financial cycle with lower leverage and smaller-sized balance sheets the credit channel of monetary is likely to be less effective – particularly at the effective lower bound.²¹ Given evidence on longer-duration financial cycles – potentially leading to a divergence of financial and business cycles – it may occur that monetary policy effectiveness is different for similar states of the business cycle, as the financial cycle can be in a different phase. Using a similar line of reasoning, the effectiveness of macroprudential policies may also vary over the financial and business cycle. Third, robust policy can benefit from combining information on leverage from the joint flow of credit and asset prices, which better capture the build-up of systemic risk than the modelling of individual indicators or the Basel III credit-to-GDP gap. This is especially evident in a real-time setting, where robust prediction of crisis probability is essential.

Paths for future research are ample. Extensions could range from deepening the interplay of financial cycles across countries – such as the exact lead-lag relation in spillovers – and their interaction with macroeconomic policies. Just as important, theoretical work is clearly needed to complement these empirical estimates and explain country-specific financial cycles, but also diverging financial and business cycle phases to inform the best use of policies.

Appendix A

A.1. Data

The financial cycle data are from the Bank of International Settlements (BIS), Datastream (DS), Global Financial Data (GFD), Haver Analytics (HA), or the OECD. All series are measured in real terms and deflated with the respective country consumer price index (CPI). Seasonal adjustment is carried out using X-12.

Total Credit: Total credit is taken from the BIS data set and reflects total loans and debt securities provided by domestic banks, all other sectors of the economy, and non-residents to the private non-financial sector (non-financial corporations, households, and non-profit institutions serving households) adjusted for breaks.

House prices: This measure reflects real residential property prices from the BIS. Italy's house price series starts in 1970Q3.

²¹ For evidence on a similar mechanism, please see [Eickmeier et al. \(2016\)](#).

Equity prices: We employ the index from OECD main economic indicators, except in the case of the US. For the US, we opt for the S&P500 index instead of the OECD main indicator, as this is more standard in academic research. The Datastream code is S&PCOMP.

Corporate bond yields: Corporate bond yields are downloaded from GFD. For Canada, the long-term corporate bond yields are employed until 2006Q1 (Source: Bank of Canada), whereas for periods after 2006Q1 we use the Bank of America Merrill Lynch Canada Corporate Effective Yield from HA. For Germany, we use yields on debt securities outstanding issued by residents provided by the Bundesbank. For France, we take the first-class private bonds average yields provided by the Banque de France. For Italy, the average corporate bond yield is provided by Banca d'Italia. For Japan, data are from The Economist. From November 2011, the yield on Nomura Securities bonds is used. For the UK, data are taken from an index of corporate bond yields as calculated by the Financial Times. From November 2011, the EIB 2028 bond is used. For the US, we use Moody's Corporate BAA yield. Moody's tries to include bonds with remaining maturities as close as possible to 30 years. Moody's drops bonds if the remaining life falls below 20 years, if the bond is susceptible to redemption, or if the rating changes.

Inflation: CPI is taken from the OECD main economic indicators database.

Business cycle indicators: The business cycle data set builds on [Ohanian and Raffo \(2012\)](#). We build on the dataset in the following sense: We obtain an updated dataset covering the time period until the end of 2016 (<http://www.andrearaffo.net/raffo/Research.html>, last accessed 1/07/19). We extend this dataset until 2018Q4 using the sources mentioned in the study. The data by [Ohanian and Raffo \(2012\)](#) provide real GDP, real private consumption, and real gross fixed capital formation from the OECD Economic Outlook database. Furthermore, the study derives its own quarterly dataset on hours worked, which has the advantage of being measured consistently across countries. The dataset combines a variety of international sources (national statistical offices, establishment surveys, and household surveys). First, an OLS regression is used to backcast the official series with the available surveys. Second, the derived quarterly series of hours worked is adjusted such that it conforms with the respective annual series on hours worked per worker of the Total Economy Database (TED). The authors do so by minimising the weighted squared deviations of the previously constructed series and the final series of hours worked, where the final series is constrained to match the yearly value of the TED series.²²

A.2. Statistical properties of financial cycle indicators relative to business cycle indicators

The nascent literature on financial cycles indicates that the medium-term component is more relevant relative to the one in output. In this vein, we compare three summary statistics of financial cycle indicators with those of business cycle indicators, dividing each variable's cycle (2–200 quarters) into a short- (2–32 quarters) and a medium- and long-term (32–200) frequency component. We consider standard deviation, autocorrelation, and cross-correlation between indicators, akin to [Comin and Gertler \(2006\)](#).²³ For cross-correlation between indicators we use credit as the base variable because of its prominent role as a financial cycle indicator; output is chosen for business cycle indicators. Results for the US are reported in [Table 3](#) and [Table 4](#). The findings for all other countries are collected in [Table 5](#) and [Table 6](#).

Were the longer-term components (i.e. the medium- and long-term components) of financial cycle indicators more relevant than for business cycle indicators, we would expect three conditions to hold. First, the *variance* of the longer-frequency component (32–200 quarters) relative to the overall variance would be larger for financial cycle indicators than for business cycle indicators. Second, the *persistence* of financial cycle indicators would be located more in the longer-frequency component compared with business cycle indicators. Third, the highest *correlation* of other indicators with credit (as an anchor variable of the financial cycle) can be attributed rather to the longer-term component when compared with the highest correlation of other indicators with output (as an anchor variable of the business cycle).

For the narrow financial cycle measures (credit and house prices) and with respect to the first property above (*variance*), we indeed find evidence for all G-7 countries that the variance of the longer-frequency component relative to the overall variance is larger for financial cycle than for business cycle indicators. The weakest evidence of this difference is found for Canada and house prices. With respect to the second property above (*persistence*), the results indicate that the longer-term cycle is far more persistent than when considering only a high-frequency component for narrow financial cycle indicators compared with business cycle indicators. In the case of the US, the first-order autocorrelation increases from 0.48 to 0.76 for credit and 0.54 to 0.81 for house prices. In contrast, the increase of business cycle indicators is at most from 0.20 to 0.32, for consumption. For house prices in Canada, the increase in persistence is of similar magnitude as for output. For the third property above (*cross-correlation*), for the narrow financial cycle indicators, the longer-term cycle is more strongly correlated than the short-frequency component for both credit and house prices. Furthermore, we find that for all G-7 countries maximum correlation in the medium/long-term is contemporaneous. Business cycle indicators have strong correlations with output that are largely impervious to adding longer-term cycles. Similarly, all business cycle indicators share highest comovement contemporaneously.

The results for narrow financial cycles to some extent carry over when considering a broad financial cycle, i.e. also including both equity and bond prices. For *equity prices*, though, only a small percentage of *variation* is located in the longer-frequency component when compared with shorter frequencies. At the same time, the absolute variation located at longer-term frequencies is by far the largest compared with all other indicators – at 1.98 for the US. The characteristics

²² The TED has been developed by the Conference Board and the Groningen Growth and Development Centre.

²³ Note that [Comin and Gertler \(2006\)](#) analyse growth cycles and not cycles in growth rates.

Table 3

US: Descriptive statistics – real growth rates.

Financial cycle	Standard deviations			First-order autocorrelations	
	Medium/long-term cycle 2–200	Short-frequency component 2–32	Medium/long-frequency component 32–200	Medium/long-term cycle 2–200	Short-frequency component 2–32
Credit (Δcr)	0.94	0.66	0.66	0.74	0.48
House prices (Δp_h)	1.62	1.04	1.22	0.81	0.54
Equity prices (Δp_e)	7.71	7.46	1.91	0.04	–0.03
Bond prices (Δp_b)	0.86	0.70	0.47	0.64	0.47
Business cycle					
Output (Δq)	0.79	0.75	0.24	0.30	0.22
Consumption (Δco)	0.63	0.58	0.25	0.32	0.20
Investment (Δi)	1.98	1.83	0.74	0.48	0.39
Hours worked (Δh)	0.35	0.35	0.06	–0.14	–0.17

Notes: Bold numbers indicate significance at least at the 10% level. Statistics are derived using HAC standard errors.

Table 4

US: Maximum absolute correlation with credit and output – leading or lagging up to 4 quarters.

Financial cycle	Credit				Output			
	Medium/long-term cycle 2–200		Short-frequency component 2–32		Medium/long-term cycle 2–200		Short-frequency component 2–32	
Credit (Δcr)	–	(–)	–	(–)	0.40	(2)	0.33	(2)
House prices (Δp_h)	0.47	(0)	0.34	(0)	0.39	(–1)	0.36	(–1)
Equity prices (Δp_e)	0.19	(–3)	0.17	(–3)	0.33	(–1)	0.30	(–1)
Bond prices (Δp_b)	0.42	(0)	0.48	(0)	0.40	(–2)	0.39	(–2)
Business cycle								
Output (Δq)	0.40	(–2)	0.33	(–2)	–	(–)	–	(–)
Consumption (Δco)	0.55	(0)	0.50	(0)	0.66	(0)	0.62	(0)
Investment (Δi)	0.43	(–2)	0.42	(0)	0.76	(0)	0.74	(0)
Hours worked (Δh)	0.16	(–4)	0.17	(–4)	0.43	(0)	0.44	(0)

Notes: Numbers in brackets denote lead (+) or lag (–) of credit or output relative to the respective variable.

of *persistence* in equity prices is similar to that of business cycle indicators. In the case of the US, this is particularly clear, with no significant first-order autocorrelation in either the high-frequency component or the longer-term cycle. The cross-correlation with credit is mostly located in the high-frequency component. Interestingly, however, equity prices seem to be leading credit by several quarters. *Bond prices*, in contrast, share similar features with credit and house prices. That is, an important share of *variance* is located at the longer-frequency component and cycles in the longer-frequency band are highly *persistent*. The results, however, are rather mixed with regard to the *cross-correlation* with credit. Both the longer-term cycle and the short-frequency component show high correlation. In most cases this is contemporaneous.

In sum, credit and house prices (and also bond prices, to some extent) share characteristics (in their variance, persistence and correlations) which are distinct from business cycle indicators. For these indicators, the highest correlation or cohesion is contemporaneous, similar to the analogous finding for business cycle indicators. Equity prices represent an exception to these regularities among financial cycle indicators. While equity prices are marked by highest variance in the longer-frequency component, this is of relatively low importance in capturing overall variance for this volatile series. Furthermore, equity prices appear to lead credit. Business cycle indicators also seem to be related to credit at longer-term frequencies, however, with no systematic lead-lag relationship across our set of countries.

A.3. Power cohesion

Proof of Proposition 1. The cross-spectrum may be written as $s_{x_i x_j}(\omega) = c_{x_i x_j}(\omega) - i q_{x_i x_j}(\omega)$, where the real part is captured by the co-spectrum $c_{x_i x_j}(\omega)$ and the imaginary part by the quadrature spectrum $q_{x_i x_j}(\omega)$. The real part describes the covariance in-phase, whereas the imaginary part describes the one out-of-phase.²⁴ Using the modulus operation in Eq. (1), we can see that

$$|f_{x_i x_j}(\omega)| = \frac{|s_{x_i x_j}(\omega)|}{\sigma_{x_i} \sigma_{x_j}} = \frac{\sqrt{c_{x_i x_j}^2(\omega) + q_{x_i x_j}^2(\omega)}}{\sigma_{x_i} \sigma_{x_j}}. \quad (8)$$

²⁴ Out-of-phase refers to cycles that are not contemporaneous. As a possible example, consider two cosine waves that are identical, except that one is shifted by $\pi/2$. The usual correlation coefficient between the two series is zero, while the underlying cycles are the same.

Table 5

Descriptive statistics – real growth rates.

Country	Variables	Standard deviations			First-order autocorrelations	
		Med./long-term cycle 2–200	Short-frequency component 2–32	Med./long-frequency component 32–200	Med./long-term cycle 2–200	Short-frequency component 2–32
Canada	<i>Financial cycle:</i>					
	Credit (Δcr)	1.00	0.80	0.58	0.50	0.23
	House prices (Δp_h)	2.41	2.23	0.91	0.39	0.29
	Equity prices (Δp_e)	7.21	7.04	1.48	0.26	0.23
	Bond prices (Δp_b)	0.79	0.61	0.47	0.63	0.38
	<i>Business cycle:</i>					
	Output (Δq)	0.80	0.74	0.29	0.37	0.26
	Consumption (Δco)	0.75	0.68	0.31	0.09	–0.11
	Investment (Δi)	2.24	2.10	0.69	0.35	0.27
	Hours worked (Δh)	0.40	0.39	0.08	–0.08	–0.13
Germany	<i>Financial cycle:</i>					
	Credit (Δcr)	0.74	0.63	0.37	0.32	0.08
	House prices (Δp_h)	0.82	0.69	0.40	0.43	0.20
	Equity prices (Δp_e)	8.04	7.71	2.12	0.35	0.30
	Bond prices (Δp_b)	0.64	0.53	0.34	0.48	0.24
	<i>Business cycle:</i>					
	Output (Δq)	0.94	0.92	0.20	0.11	0.07
	Consumption (Δco)	0.91	0.87	0.22	–0.23	–0.31
	Investment (Δi)	2.55	2.47	0.58	–0.13	–0.21
	Hours worked (Δh)	0.61	0.61	0.06	–0.26	–0.28
France	<i>Financial cycle:</i>					
	Credit (Δcr)	0.94	0.78	0.50	0.48	0.23
	House prices (Δp_h)	1.25	0.77	0.98	0.81	0.50
	Equity prices (Δp_e)	8.71	8.37	2.35	0.33	0.27
	Bond prices (Δp_b)	0.86	0.61	0.58	0.70	0.41
	<i>Business cycle:</i>					
	Output (Δq)	0.53	0.47	0.23	0.49	0.37
	Consumption (Δco)	0.59	0.54	0.23	0.03	–0.15
	Investment (Δi)	1.16	0.94	0.63	0.61	0.42
	Hours worked (Δh)	0.34	0.31	0.12	0.79	0.76
Italy	<i>Financial cycle:</i>					
	Credit (Δcr)	1.41	1.07	0.89	0.46	0.08
	House prices (Δp_h)	1.85	1.45	1.08	0.81	0.69
	Equity prices (Δp_e)	10.38	9.79	3.36	0.39	0.31
	Bond prices (Δp_b)	1.25	0.94	0.78	0.69	0.45
	<i>Business cycle:</i>					
	Output (Δq)	0.78	0.74	0.23	0.47	0.42
	Consumption (Δco)	0.68	0.62	0.29	0.34	0.19
	Investment (Δi)	1.69	1.53	0.70	0.26	0.10
	Hours worked (Δh)	0.55	0.54	0.11	–0.28	–0.34
Japan	<i>Financial cycle:</i>					
	Credit (Δcr)	1.17	0.90	0.68	0.55	0.25
	House prices (Δp_h)	1.56	1.13	0.95	0.77	0.57
	Equity prices (Δp_e)	8.34	7.81	2.80	0.38	0.29
	Bond prices (Δp_b)	1.02	0.80	0.58	0.57	0.30
	<i>Business cycle:</i>					
	Output (Δq)	1.03	1.00	0.25	0.10	0.03
	Consumption (Δco)	1.13	1.10	0.20	–0.24	–0.30
	Investment (Δi)	1.98	1.82	0.71	0.13	–0.02
	Hours worked (Δh)	0.90	0.88	0.15	–0.41	–0.45
UK	<i>Financial cycle:</i>					
	Credit (Δcr)	1.95	1.62	1.08	0.39	0.12
	House prices (Δp_h)	2.62	2.12	1.42	0.78	0.67
	Equity prices (Δp_e)	7.84	7.59	1.92	0.27	0.22
	Bond prices (Δp_b)	1.15	0.91	0.68	0.59	0.34
	<i>Business cycle:</i>					
	Output (Δq)	0.91	0.85	0.30	0.20	0.08
	Consumption (Δco)	1.02	0.92	0.41	0.12	–0.07
	Investment (Δi)	2.81	2.70	0.72	0.03	–0.05
	Hours worked (Δh)	0.40	0.39	0.10	–0.21	–0.28

Notes: Bold numbers indicate significance at the 10% level. Statistics are derived using HAC standard errors. Med. refers to medium.

Table 6

Maximum absolute correlation with domestic credit and output – leading or lagging up to 4 quarters.

Country	Variable	Domestic credit				Domestic output			
		Med./long-term cycle 2–200		Short-frequency component 2–32		Med./long-term cycle 2–200		Short-frequency component 2–32	
Canada	<i>Financial cycle:</i>								
	Credit (Δcr)	–	(–)	–	(–)	0.34	(2)	0.28	(2)
	House prices (Δp_h)	0.31	(0)	0.22	(0)	0.29	(–1)	0.26	(–1)
	Equity prices (Δp_e)	–0.21	(4)	0.23	(–4)	0.39	(–1)	0.40	(–1)
	Bond prices (Δp_b)	–0.30	(4)	–0.24	(4)	0.33	(–3)	0.30	(–3)
	<i>Business cycle:</i>								
	Output (Δq)	0.34	(–2)	0.28	(–2)	–	(–)	–	(–)
	Consumption (Δco)	0.39	(–3)	0.25	(–3)	0.53	(0)	0.46	(0)
	Investment (Δi)	0.30	(–3)	0.23	(–1)	0.51	(0)	0.47	(0)
	Hours worked (Δh)	0.15	(–3)	0.15	(–3)	0.24	(0)	0.23	(0)
Germany	<i>Financial cycle:</i>								
	Credit (Δcr)	–	(–)	–	(–)	0.17	(0)	0.13	(–4)
	House prices (Δp_h)	0.31	(0)	0.25	(0)	0.19	(1)	0.14	(1)
	Equity prices (Δp_e)	–0.13	(4)	–0.11	(4)	0.22	(–2)	0.24	(–1)
	Bond prices (Δp_b)	–0.25	(4)	–0.21	(4)	0.23	(–3)	0.24	(–3)
	<i>Business cycle:</i>								
	Output (Δq)	0.17	(0)	0.13	(4)	–	(–)	–	(–)
	Consumption (Δco)	0.24	(–4)	0.18	(–4)	0.49	(0)	0.46	(0)
	Investment (Δi)	0.13	(0)	0.10	(0)	0.73	(0)	0.72	(0)
	Hours worked (Δh)	–0.13	(–1)	–0.14	(–1)	0.36	(0)	0.38	(0)
France	<i>Financial cycle:</i>								
	Credit (Δcr)	–	(–)	–	(–)	0.31	(2)	0.15	(3)
	House prices (Δp_h)	0.26	(0)	0.10	(0)	0.40	(0)	0.34	(0)
	Equity prices (Δp_e)	0.13	(–4)	0.16	(1)	0.32	(–1)	0.33	(–1)
	Bond prices (Δp_b)	0.35	(0)	0.43	(0)	0.33	(–4)	0.31	(–4)
	<i>Business cycle:</i>								
	Output (Δq)	0.31	(–2)	0.15	(–3)	–	(–)	–	(–)
	Consumption (Δco)	0.28	(1)	0.17	(1)	0.55	(0)	0.46	(0)
	Investment (Δi)	0.37	(–2)	0.14	(–3)	0.76	(0)	0.72	(0)
	Hours worked (Δh)	–0.07	(4)	–0.10	(–3)	–0.22	(4)	–0.23	(4)
Italy	<i>Financial cycle:</i>								
	Credit (Δcr)	–	(–)	–	(–)	0.15	(2)	0.12	(2)
	House prices (Δp_h)	0.27	(0)	–0.15	(–3)	0.27	(2)	–0.33	(–4)
	Equity prices (Δp_e)	–0.09	(3)	0.15	(2)	0.29	(–1)	0.28	(–1)
	Bond prices (Δp_b)	0.35	(0)	0.30	(0)	0.30	(–3)	0.35	(–3)
	<i>Business cycle:</i>								
	Output (Δq)	0.15	(–2)	0.12	(–2)	–	(–)	–	(–)
	Consumption (Δco)	0.21	(–3)	–0.17	(4)	0.49	(0)	0.44	(0)
	Investment (Δi)	0.25	(–4)	0.15	(–2)	0.58	(0)	0.54	(0)
	Hours worked (Δh)	–0.12	(1)	–0.10	(1)	0.22	(0)	0.23	(0)
Japan	<i>Financial cycle:</i>								
	Credit (Δcr)	–	(–)	–	(–)	0.35	(0)	0.30	(0)
	House prices (Δp_h)	0.68	(0)	0.56	(0)	0.38	(0)	0.39	(0)
	Equity prices (Δp_e)	0.26	(0)	0.23	(0)	0.28	(–1)	0.24	(–1)
	Bond prices (Δp_b)	0.55	(0)	0.73	(0)	0.26	(0)	0.34	(0)
	<i>Business cycle:</i>								
	Output (Δq)	0.35	(0)	0.30	(0)	–	(–)	–	(–)
	Consumption (Δco)	0.43	(0)	0.44	(0)	0.71	(0)	0.71	(0)
	Investment (Δi)	0.41	(0)	0.31	(0)	0.66	(0)	0.63	(0)
	Hours worked (Δh)	0.12	(–4)	0.15	(–1)	0.19	(0)	0.19	(0)
UK	<i>Financial cycle:</i>								
	Credit (Δcr)	–	(–)	–	(–)	0.27	(0)	0.22	(0)
	House prices (Δp_h)	0.46	(0)	0.31	(0)	0.50	(0)	0.43	(0)
	Equity prices (Δp_e)	0.33	(–3)	0.33	(–3)	0.35	(–3)	0.34	(–3)
	Bond prices (Δp_b)	0.41	(0)	0.39	(0)	0.38	(0)	0.35	(0)
	<i>Business cycle:</i>								
	Output (Δq)	0.27	(0)	0.22	(0)	–	(–)	–	(–)
	Consumption (Δco)	0.34	(0)	0.23	(0)	0.67	(0)	0.62	(0)
	Investment (Δi)	0.24	(–4)	0.15	(–4)	0.39	(0)	0.34	(0)
	Hours worked (Δh)	0.16	(–3)	0.14	(–3)	0.22	(0)	0.19	(0)

Notes: Numbers in brackets denote lead (+) or lag (–) of credit or output relative to the respective variable. Med. refers to medium.

This implies that, using the PCoh methodology, both covariances (in-phase and out-of-phase) are preserved. $\square$

Proof of Proposition 2. The cross-spectrum integrates to the unconditional covariance between $x_{i,t}$ and $x_{j,t}$ (see e.g. Hamilton (1994, p. 271)),

$$\int_{-\pi}^{\pi} s_{x_i x_j}(\omega) d\omega = \text{Cov}[x_{i,t}, x_{j,t}]. \quad (9)$$

From this it follows that

$$-1 \leq \int_{-\pi}^{\pi} f_{x_i x_j}(\omega) d\omega \leq 1. \quad (10)$$

Further, taking the absolute value of the normalised cross-spectrum, i.e. $|f_{x_i x_j}(\omega)|$ and integrating it from $-\pi$ to π leads to the support

$$0 \leq \int_{-\pi}^{\pi} |f_{x_i x_j}(\omega)| d\omega \leq 1. \quad (11)$$

The upper bound of 1 in Eq. (11) holds as the covariance in-phase and out-of-phase cannot exceed the product of the standard deviations of both variables. $\square$

Proof of Proposition 3. To better understand our proposed measure, we contrast power cohesion with cohesion derived from a comparable measure based on the commonly used measure of squared coherency.

Let squared coherency cohesion be denoted by SCoh. It is defined as

$$\text{SCoh}_X(\omega) = \frac{1}{(M-1)M} \sum_{i \neq j} \frac{|s_{x_i x_j}(\omega)|^2}{s_{x_i x_i}(\omega) s_{x_j x_j}(\omega)}. \quad (12)$$

A value of $\text{SCoh}(\omega)$ of one indicates linear equivalence of processes at frequency ω and zero implies linear unrelatedness at frequency ω . Note that squared coherency cohesion relates each cross-spectrum, $s_{x_i x_j}(\omega)$, to the respective auto-spectra, $s_{x_i x_i}(\omega)$ and $s_{x_j x_j}(\omega)$. In contrast, power cohesion relates each cross-spectrum to the unconditional standard deviations, σ_{x_i} and σ_{x_j} (see Eq. (1)). This implies that frequencies which do not contribute to the average covariance are mitigated using power cohesion. In contrast, they are potentially emphasised using squared coherency cohesion.

To further clarify this point, consider the following data generating process (DGP) for indicator $i \in \{1, \dots, M\}$:

$$x_{i,t} = \psi_{i,t} + \varepsilon_{i,t}, \quad (13)$$

which is characterised by a cyclical component

$$\psi_{i,t} = \alpha_{i,t} \cos(\lambda t), \quad (14)$$

where $\alpha_{i,t}$ denotes a random amplitude, which may follow a stationary autoregressive process of order one. Furthermore, λ represents the cycle frequency.²⁵ The idiosyncratic component, $\varepsilon_{i,t}$, is white noise, but possibly contemporaneously related across indicators, i.e. with $\varepsilon_{j,t}$.

A common cycle amongst a set of indicators is present if $\alpha_{i,t} = \gamma_i \alpha_t$, where γ_i is assumed to be some nonzero constant. Then, writing $\psi_t = \alpha_t \cos(\lambda t)$, $\gamma' = (\gamma_1, \dots, \gamma_M)$, $\varepsilon'_t = (\varepsilon_{1,t}, \dots, \varepsilon_{M,t})$, and stacking equations yields

$$X_t = \gamma \psi_t + \varepsilon_t. \quad (15)$$

Since ψ_t is uncorrelated with ε_t for all t and k , it follows that the spectrum of X_t is the sum of the two spectra of ψ_t and ε_t , i.e.

$$S_X(\omega) = \gamma S_\psi(\omega) \gamma' + S_\varepsilon(\omega). \quad (16)$$

For this DGP SCoh can be written as

$$\text{SCoh}_X(\omega) = \frac{1}{(M-1)M} \sum_{i \neq j} \frac{|s_{x_i x_j}(\omega)|^2}{s_{x_i x_i}(\omega) s_{x_j x_j}(\omega)} \quad (17)$$

²⁵ A more realistic DGP would also include a phase shift of the cosine wave depending on i and t ; however, as both PCoh as well as SCoh evaluate similarities of cycles irrespective of their phase, we exclude this case for ease of exposition.

$$= \frac{1}{(M-1)M} \sum_{i \neq j} \frac{\left(\gamma_i \gamma_j s_\psi(\omega) + s_{e_i e_j}(\omega) \right)^2}{\left(\gamma_i^2 s_\psi(\omega) + s_{e_i e_i}(\omega) \right) \left(\gamma_j^2 s_\psi(\omega) + s_{e_j e_j}(\omega) \right)} \quad (18)$$

To illustrate the difference between power cohesion and squared coherence cohesion, consider two cases: (1) The absence of the white noise processes and (2) the case of perfectly correlated white noise.

In the first case SCoh becomes:

$$\text{SCoh}_X(\omega) = \frac{1}{(M-1)M} \sum_{i \neq j} \frac{\gamma_i^2 \gamma_j^2 s_\psi^2(\omega)}{\gamma_i^2 \gamma_j^2 s_\psi^2(\omega)} = 1 \quad (19)$$

The last equality holds because – due to the random amplitude $\alpha_t - s_\psi(\omega)$ has nonzero power at all frequencies. Thus, in this case SCoh cannot be used to detect the common cycle, as SCoh indicates a one for all frequencies. In contrast, PCoh detects the common cycle, since it only considers the normalised cross-spectra and not the autospectra.

In the second case SCoh becomes:

$$\text{SCoh}_X(\omega) = \frac{1}{(M-1)M} \sum_{i \neq j} \frac{\left(\gamma_i \gamma_j s_\psi(\omega) + s_e(\omega) \right)^2}{\left(\gamma_i^2 s_\psi(\omega) + s_e(\omega) \right) \left(\gamma_j^2 s_\psi(\omega) + s_e(\omega) \right)}, \quad (20)$$

since $s_{e_i e_j}(\omega) = s_{e_i e_i}(\omega) = s_e(\omega)$. Here, a trade-off between variances becomes apparent. If the white noise process has far greater variance contribution at each frequency than $\psi_{i,t}$, clearly the common cycle of interest would not be apparent using either PCoh or SCoh. Abstracting from this extreme case, for instance, given similar or smaller variance of the white noise process relative to the common cycle of interest, PCoh would detect the common cycle, but SCoh would not, as can be shown by

$$\lim_{s_\psi(\omega \neq \lambda) \rightarrow 0} \text{SCoh}_X(\omega \neq \lambda) = 1, \quad (21)$$

where, again, λ is the frequency of the common cycle of interest. Again this is due to the fact that the cross-spectrum is related to the auto-spectra. Thus, even in situations in which the white noise spectrum makes a relatively small contribution to the overall variance of the process, but is highly correlated, SCoh does not capture a common cycle.

This result implies that if another common cycle with lower variance contribution is present, it would receive the same importance as the cycle with greater variance contribution using SCoh. In contrast, PCoh reports cycles relative to their variance contribution.

Finally, note that this type of misleading conclusion can also arise if spectra are close to zero, which may follow as a consequence of overdifferencing or seasonal adjustment. $\square$

A.4. Empirical issues

We estimate the (cross-) spectral densities through cross-correlations. Further, a Parzen window with a bandwidth of $\sqrt{T} \cdot 8$ is used, which is larger than the bandwidth used in Aikman et al. (2015) and emphasises our aim to have relatively unbiased spectral density estimates, though at the cost of a higher variance. As is common, we only regard the frequency window 0 to π , which suffices to determine the cyclical properties of variables when considering the absolute value of the spectral densities.

Confidence intervals for spectral densities are bootstrapped (see, for instance, Fig. 2). As suggested in Croux et al. (2001) and described in Franke and Härdle (1992) as well as in Berkowitz and Diebold (1998), we bootstrap the individual (cross-) spectra, rather than the time series. For each cross-spectrum we resample from a χ^2_2 distribution using 5000 replications and use a Gaussian kernel to smooth each periodogram.²⁶

We propose endogenously selecting a frequency window around the peak by determining the maximum (ω_1) and minimum (ω_2) cycle lengths ($\omega_2, \omega_1 \in [2\pi/200, 2\pi/5] \wedge \omega_2 \geq \omega_1$) through

$$\min_{\omega_2 - \omega_1} \frac{\int_{\omega_1}^{\omega_2} \text{PCoh}_X(\omega) d\omega}{\int_{2\pi/5}^{2\pi/200} \text{PCoh}_X(\omega) d\omega} \geq p. \quad (22)$$

where p is defined to be in the range [0,1], such that the frequency band contains $p \cdot 100\%$ of common variation excluding variation with a frequency below 1.25 and above 50 years.²⁷ As indicated, we set $p = 0.67$. By finding the maximum and min-

²⁶ This bootstrapping procedure assumes independence of cross-spectra. However, to the best of the authors' knowledge, research in this area is still ongoing and no solution for this exact problem has been proposed.

²⁷ An unrestricted range would select very short frequencies, which can be related to the high amplification that occurs using the quarter-on-quarter filter.

imum cycle length that minimises the distance $\omega_2 - \omega_1$, we are assured of picking the most important common fluctuations across indicators around the peak; i.e. the densest area of PCoh around the peak.

A.5. Cross-spectral densities and power cohesion

See Figs. 8–10.

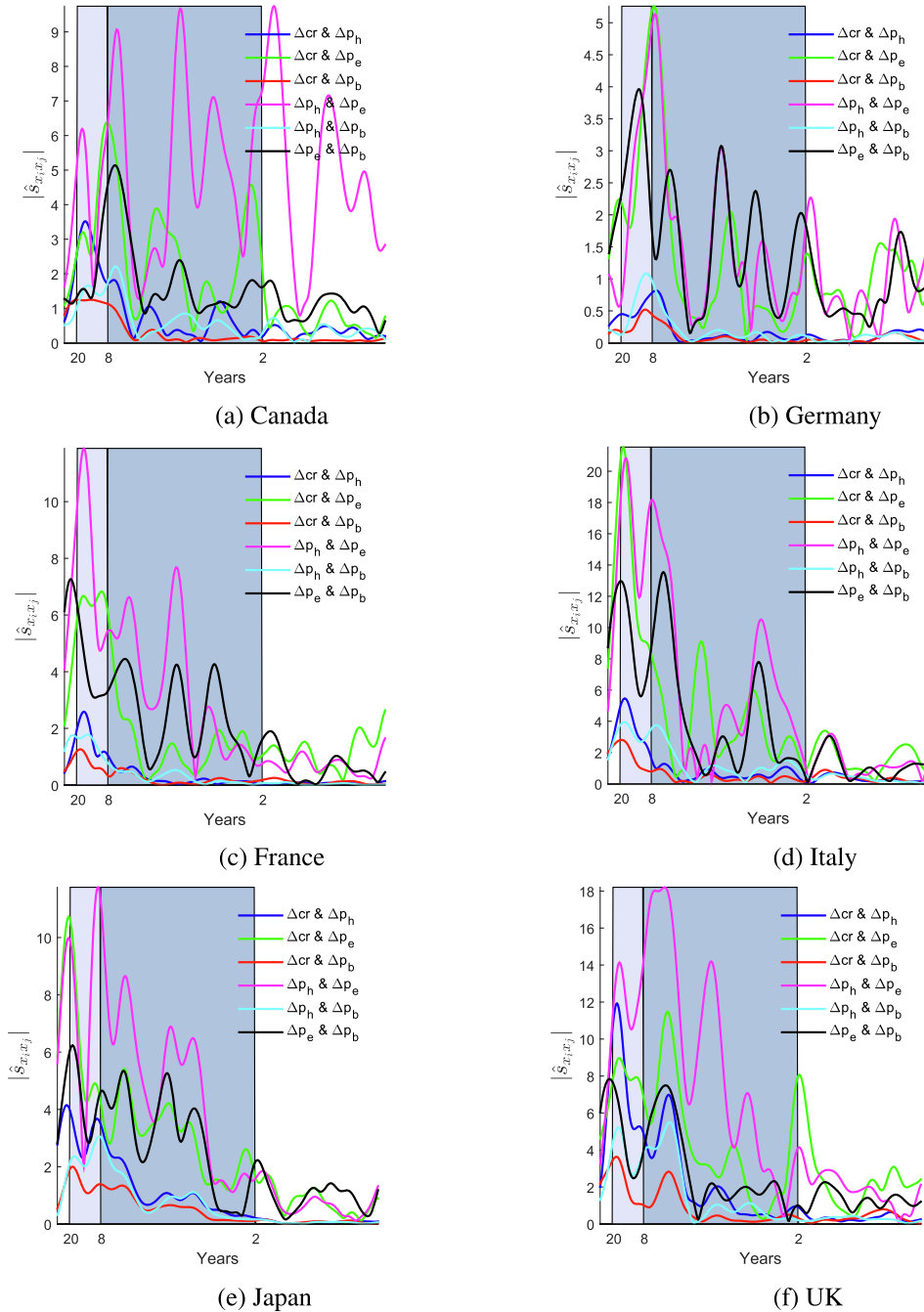


Fig. 8. Absolute cross-spectra for financial cycle indicators. *Notes:* This panel shows the absolute cross-spectra of the financial indicators. The x-axis measures the frequencies of cycles from 1.25 to 50 years. The blue area depicts business cycle frequencies, i.e. cycles with durations of 2.5–8 years and the purple area marks medium-term cycles (8–20 years). Δcr refers to percentage changes in total credit, Δp_h to percentage changes in house prices, Δp_e to percentage changes in equity prices, Δp_b to percent changes in bond prices. For figures to appear in color, please see web version of this article.

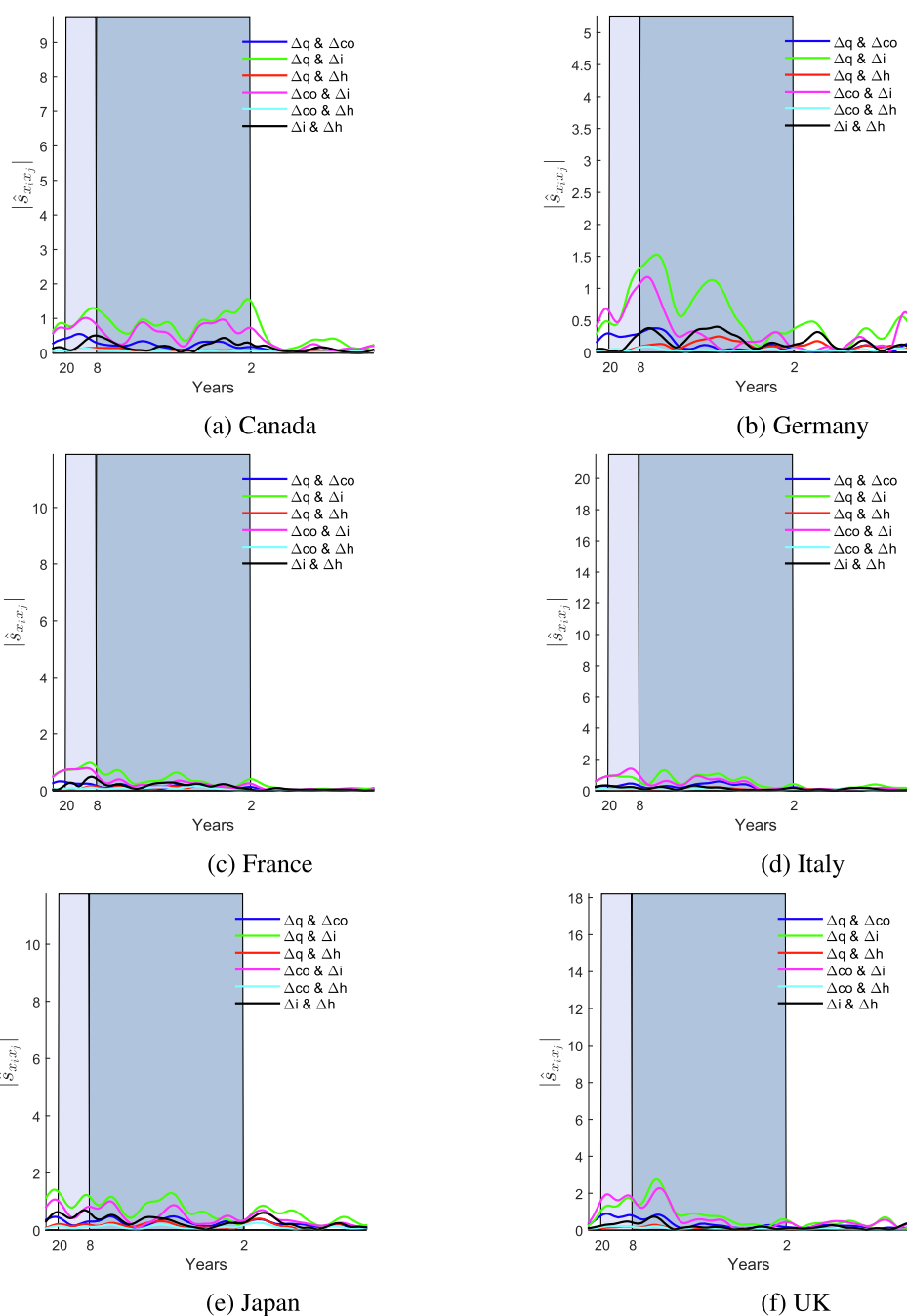


Fig. 9. Absolute cross-spectra of business cycle indicators. *Notes:* This panel shows the absolute cross-spectra of the business cycle indicators. The x-axis measures the frequencies of cycles from 1.25 to 50 years. The blue area depicts business cycle frequencies, i.e. cycles with durations of 2.5–8 years and the purple area marks medium-term cycles (8–20 years). Δq refers to percentage changes in GDP, Δco to percentage changes in consumption, Δi to percentage changes in investment, and Δh percentage changes in hours worked. For figures to appear in color, please see web version of this article.

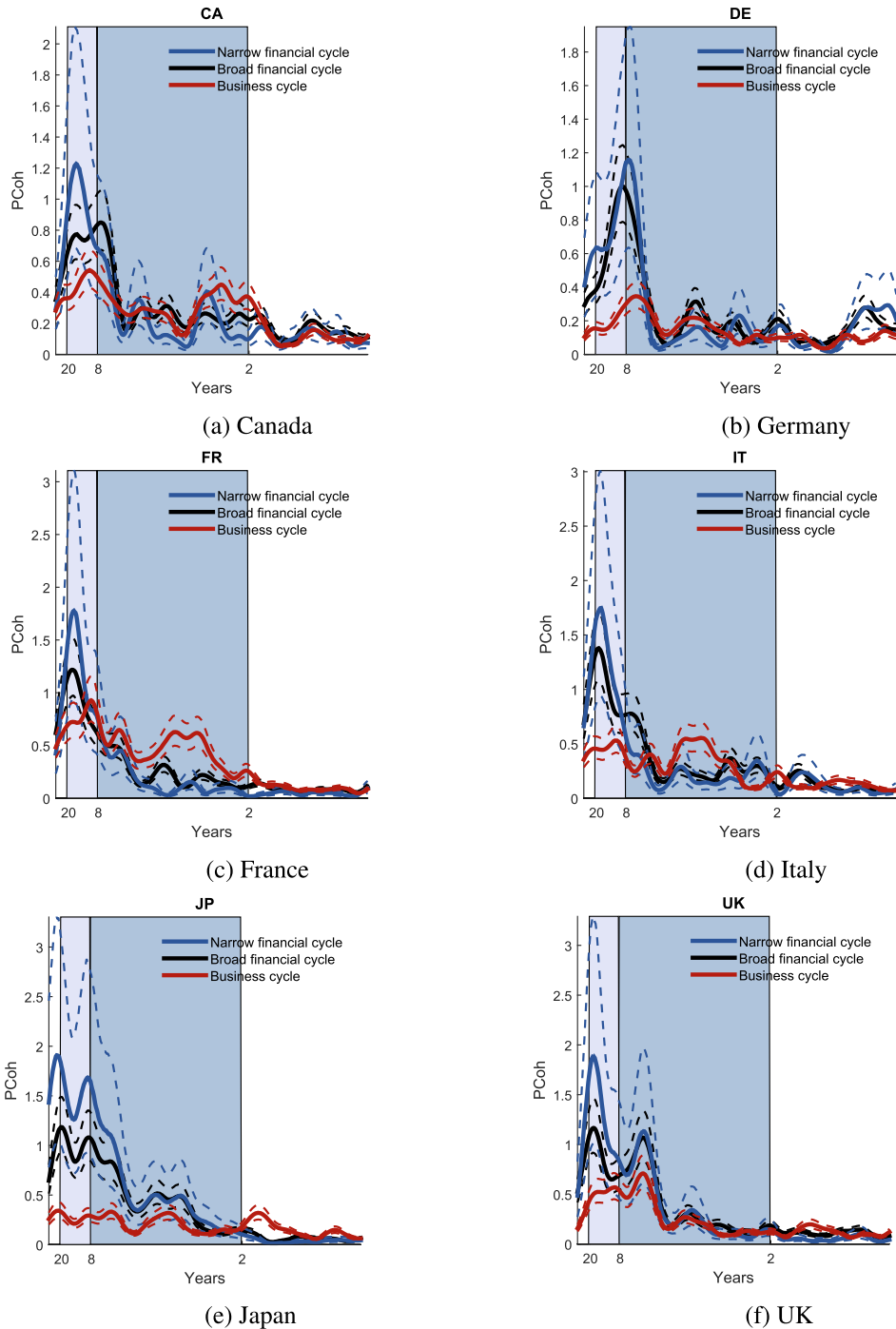


Fig. 10. Power cohesion of G-7 countries. *Notes:* This graph shows the measure power cohesion of the narrow (black lines) and broad financial cycle (blue lines) as well as the business cycle (red lines). Broad refers to the inclusion of all indicators, i.e. credit, house, equity, and bond prices, whereas narrow is defined by house prices and credit only. The dashed lines indicate the 68% bootstrapped confidence intervals as discussed in [Appendix A.4](#). The x-axis measures the frequencies of cycles from 1.25 to 50 years. The blue area depicts business cycle frequencies, i.e. cycles with durations of 2–8 years and the purple area (8–20 years) marks the frequencies that are most important for financial cycles. For figures to appear in color, please see web version of this article.

A.6. Composite financial and business cycles

See Fig. 11.

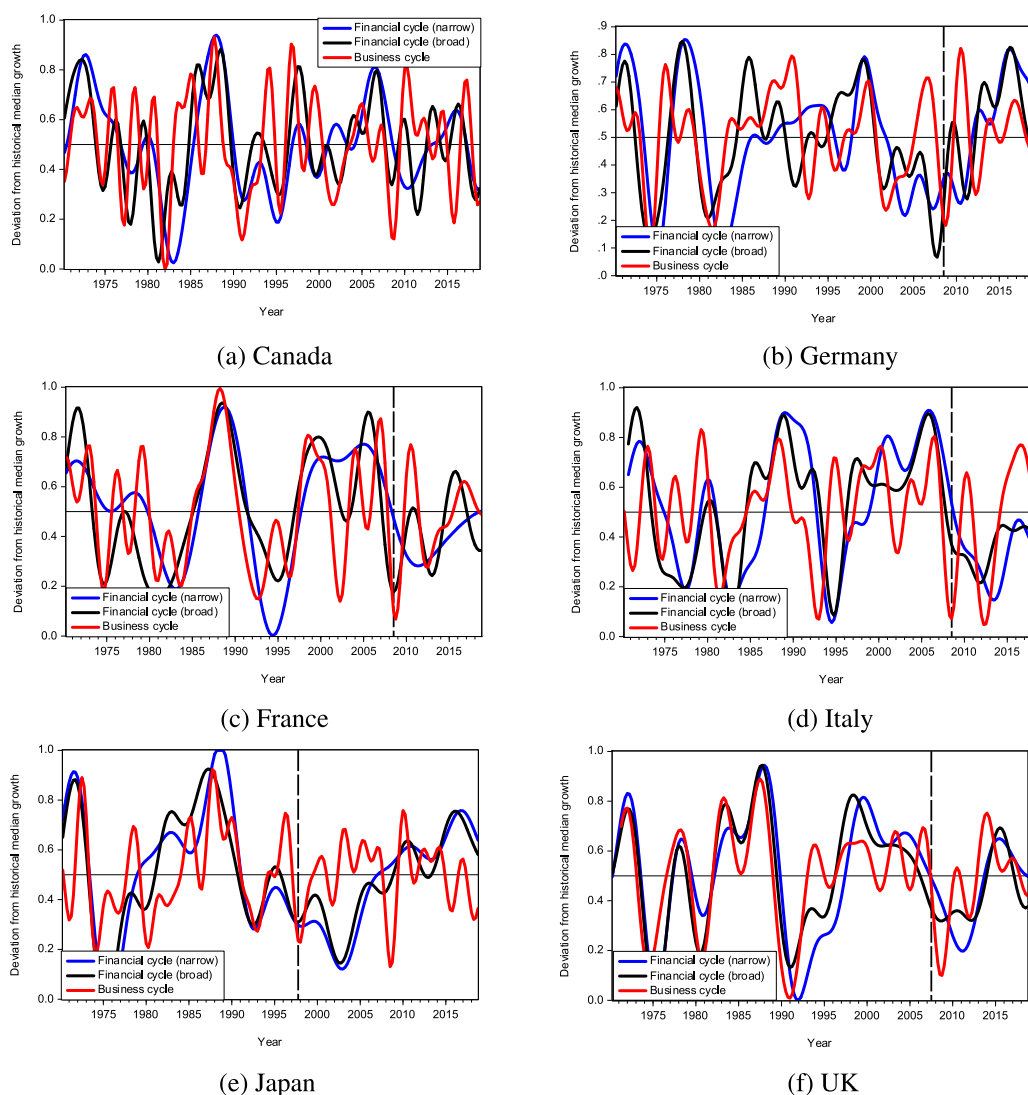


Fig. 11. Composite cycles. *Notes:* This panel shows composite financial and business cycles in standardised growth rates, where 0.5 denotes the historical median after removing a nonlinear trend; 0 is the smallest and 1 the largest growth rate observed in a country's history. Specifically, it depicts the filtered financial (narrow and broad) and business cycles. Filtering is done using the [Christiano and Fitzgerald \(2003\)](#) band pass filter employing country-specific frequency windows as described in [Table 1](#). Aggregation of standardised indicators is done using linear averaging in the case of the narrow financial cycle and using time-varying weights that emphasise directional movements between indicators for the broad financial cycle. See [Section 4.1](#) for further details. Black vertical dashed lines indicate the onset of systemic banking crises as defined by [Laeven and Valencia \(2012\)](#).

A.7. Real-time composite financial and business cycles

See Fig. 12.

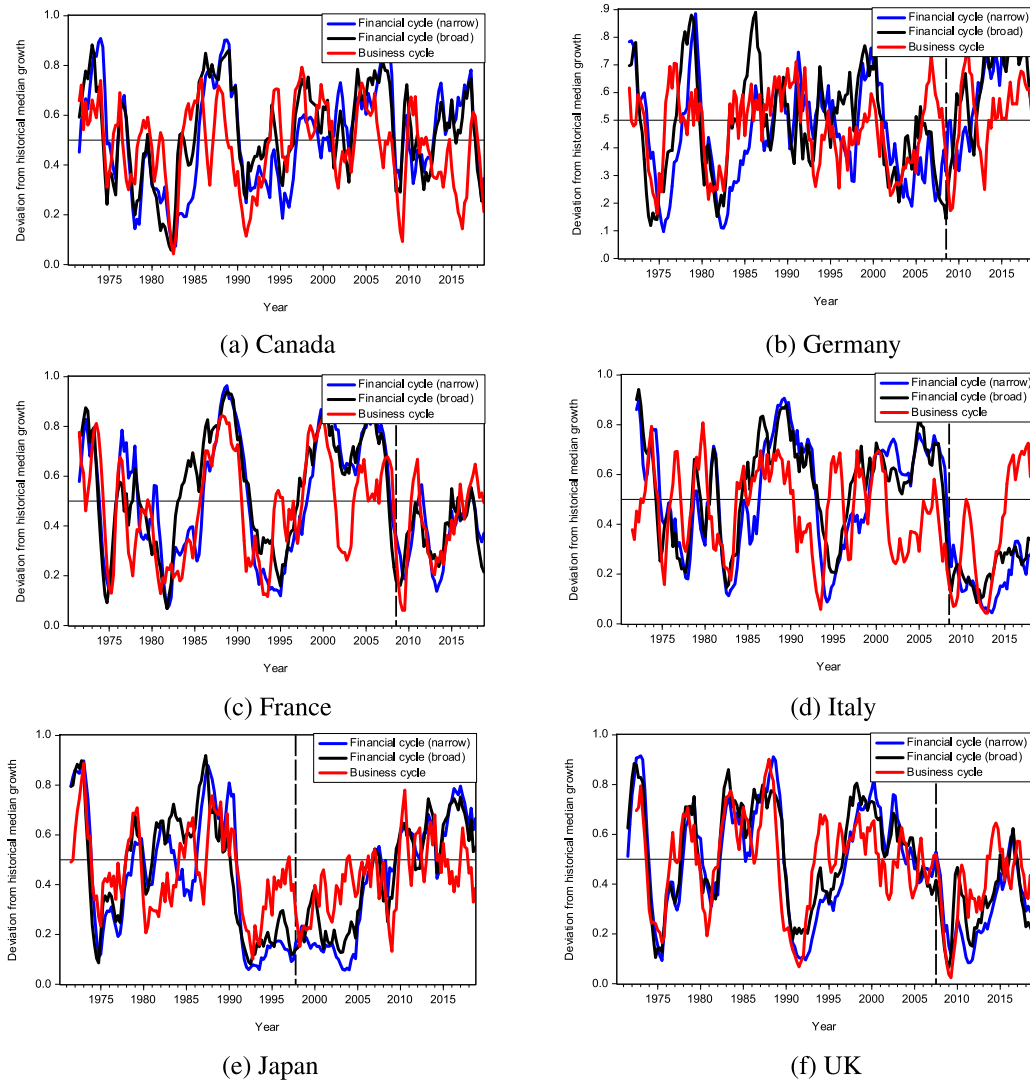


Fig. 12. Real-time composite cycles. *Notes:* The real-time composite financial and business cycles are measured in standardised growth rates, where 0.5 denotes the historical median; 0 is the smallest and 1 the largest growth rate observed in a country's history. Smoothing is done using a six-quarter one-sided moving average based on the weights of a Bartlett window. See Section 4.1 and 4.4.1 for further details. Black vertical dashed lines indicate the onset of systemic banking crises as defined by Laeven and Valencia (2012).

A.8. Signalling exercise: Robustness

See Table 7.

Table 7

Signalling exercise accounting for a post-crisis bias.

Indicators	One quarter out-of-sample											In-sample		
	Obs.	K	TP	FP	TN	FN	TI	TII	U^r	NtS	AUC	Obs.	K	AUC
Panel B: 1–4 quarters before crisis														
Financial cycle (narrow)	497	5	13	217	260	7	0.35	0.45	0.20	0.70	0.69***	1008	5	0.76**
Financial cycle (broad)	497	3	14	132	345	6	0.30	0.28	0.42	0.40	0.73***	1008	5	0.86***
Business cycle	497	3	11	213	264	9	0.45	0.45	0.10	0.81	0.58	1008	5	0.67

Table 7 (continued)

Indicators	One quarter out-of-sample											In-sample		
	Obs.	K	TP	FP	TN	FN	TI	TII	U ^r	NtS	AUC	Obs.	K	AUC
Δcr	497	6	12	117	360	8	0.40	0.25	0.35	0.41	0.62**	1008	5	0.73**
$\Delta cr \& \Delta p_h$	497	6	13	113	364	7	0.35	0.24	0.41	0.36	0.57	1008	5	0.76***
$\Delta cr, \Delta p_h, \Delta p_e, \& \Delta p_b$	497	4	10	163	314	10	0.50	0.34	0.16	0.68	0.47	1008	5	0.83***
Credit-to-GDP gap	497	1	8	72	405	12	0.60	0.15	0.25	0.38	0.53	1008	2	0.63
Panel C: 5–12 quarters before crisis														
Financial cycle (narrow)	497	2	19	138	319	21	0.53	0.30	0.17	0.64	0.62*	1008	3	0.66
Financial cycle (broad)	497	1	12	48	409	28	0.70	0.11	0.19	0.35	0.61*	1008	5	0.66
Business cycle	497	1	34	306	151	6	0.15	0.67	0.18	0.79	0.46***	1008	5	0.58
Δcr	497	2	21	63	394	19	0.48	0.14	0.39	0.26	0.61*	1008	2	0.67*
$\Delta cr \& \Delta p_h$	497	2	23	57	400	17	0.43	0.12	0.45	0.22	0.63**	1008	5	0.70***
$\Delta cr, \Delta p_h, \Delta p_e, \& \Delta p_b$	497	2	22	166	291	18	0.45	0.36	0.19	0.66	0.50*	1008	5	0.78***
Credit-to-GDP gap	497	1	30	247	210	10	0.25	0.54	0.21	0.72	0.57	1008	5	0.63

Notes: Table shows results of the out-of- and in-sample exercise as described in Section 4.4.1. "Obs." refers to observations, "K" to the number of lags, "TP" to true positives, "FP" to false positives, "TN" to true negatives, "FN" to false negatives, "TI" to Type I error, "TII" to Type II error, "U^r" to relative usefulness, "NtS" to noise-to-signal ratio, and "AUC" to area under the curve. *, **, and *** indicate that the AUC of the respective indicator is significantly different to the AUC of the credit-to-GDP gap at the 10%, 5%, and 1% significance level, using the test proposed by DeLong et al. (1988) that accounts for correlation between two curves. "-" indicates that the statistic is not defined.

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